

Canadian Content Regulations and Broadcasting: Evidence from the 1999 CanCon Expansion*

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This version: October 2, 2026

Abstract

Content quotas mandate exposure to domestic cultural goods but leave intermediaries to decide which goods receive it. I study how Canadian radio broadcasters allocated the exposure added by the 1999 expansion of the Canadian content quota, which raised the weekly floor from 30 to 35 percent and extended it to weekday daytime hours. Linking digitized Canadian radio charts for 1994–2000 to a music catalogue built from MusicBrainz and Discogs, I compare Canadian and non-Canadian artists before and after the expansion. The quarterly probability that an active Canadian artist charts falls by 1.7 percentage points, 40 percent of its baseline, while the chart slots held by Canadian artists do not change, so the mandated exposure is concentrated among fewer artists. Canadian chart runs lengthen by 3 issues, 23 percent, entirely after their peak and in the middle and lower thirds of the charts, and charting Canadian songs are older. The pre-policy relationship between Canadian chart runs and US chart success would imply an 11 percentage point increase in US charting; after the expansion this relationship weakens and the implied effect is small and imprecise. The results indicate that broadcasters met the quota by extending the life of proven songs rather than by broadening exposure to Canadian music.

JEL Classification: L82, Z11, Z18, L51, L52

Keywords: content quotas, cultural policy, radio broadcasting, intermediaries, consumption capital, music charts

*I thank Torsten Jaccard for helpful comments and discussions as my Summer Paper advisor. I also thank Sam Norris, Pierre-Loup Beauregard, Ron Yang, and Guillermo Marshall for their feedback and support, as well as the participants of the TRIOS brownbag. All errors are my own.

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1 Introduction

Industrial policy, often implicitly defined as the use of government instruments to promote activity in specific industries, has seen a surge in political popularity in recent years. In manufacturing, the economic justification for such policies relies on the presence of positive externalities which private markets may fail to internalize, such as agglomeration effects or the benefits of learning-by-doing (Juhász et al., 2024). An alternative, though less discussed, positive externality associated with consumption goods is the possibility that consumer exposure to new goods is itself a positive externality, given the network effects associated with consumer goods and the frictions consumers face in searching for new goods (Argente et al., 2026).

This externality has been considered in the cultural goods literature, where learning about own preferences and the quality of the good requires consumption (Lévy-Garboua et al., 2025; Throsby et al., 2024). As well, this externality implies that there are valuable cultural goods that are not being sold or produced because of too little access to exposure, and this can be especially true if there is home-bias in preferences. This suggests a rationale for policies that provide exposure to cultural goods, such as a content quota, funding mechanisms, or tournaments. Governments have responded to this concern through policies that increase exposure to domestic cultural goods, including content quotas. In Canada, such a quota was implemented as a 30% Canadian content mandate for weekly radio airplay in 1971, increased to 35% in 1999¹. But content quotas do not determine which goods receive the mandated exposure. Instead, they constrain the aggregate choices of the intermediaries that select what consumers see and hear. The effectiveness of the policy may therefore depend on how these intermediaries allocate the exposure they are required to provide.

This paper addresses this question in the context of Canadian radio: I study how broadcasters allocate the exposure mandated by the quota and whether that allocation generates broader exposure to Canadian music by leveraging the expansion of the content requirements in 1999. The central question is whether a content quota can overcome an exposure externality when the intermediary responsible for allocating the quota has its own strategic incentives. Understanding which artists get the extra airtime is particularly important considering a recent move in Quebec to expand the French language quota to include streaming services, and recent frictions with the US in trade talks regarding the same French language quota (Harvey, 2026; The Canadian Press, 2025).

I use music charts and crowd-sourced catalogue metadata to trace the allocation of radio exposure across Canadian and non-Canadian works before and after the January 1999 quota change². I first ask whether the additional mandated exposure changes the extensive margin of Canadian music: do new artists enter the market and release new works, and do any of these new works gain more airplay? To solve the incomplete market that consumption capital introduces, the quota has to allow commercially viable artists to accumulate consumption capital. This only happens if new artists are given access to exposure. I then characterize how exposure is distributed across artists and works: whether the additional exposure is broadly spread across

¹Australia was a pioneer in content quotas for radio broadcasting, with Canada following suit (Filimon et al., 2011). Nowadays, this type of policy is widespread around the world, with countries such as France, Portugal, and Hungary implementing similar schemes, some of them also with provisions for new artists.

²CanCon experienced several expansions from its inception in 1971 that I explain in more detail in the following sections and the Appendices. The 1999 expansion is used here mainly due to better data quality in more recent periods, but a following step is to analyze the prior expansions as these occurred over different regimes of the recording and broadcasting industries.

Canadian acts or concentrated among a smaller set, whether artists receive longer spells or reach higher peaks, and whether the quota changes the degree of rotation in Canadian music. These patterns distinguish between a quota that expands the set of Canadian music reaching consumers and one that primarily intensifies exposure to music that broadcasters already find commercially attractive. Additionally, I ask if the extra local exposure translates into increased international exposure. I use the resulting changes in aggregate broadcasters' selection and scheduling behavior to evaluate the extent to which strategic allocation mitigates the intended exposure effect of the quota.

I find that the average Canadian artist has a lower probability of appearing in a music chart in Canada after the quota expansion while the total amount of chart slots occupied by Canadian artists remains unchanged. This can only happen if each charted artist has more songs entering music charts or each song is staying in the charts for longer. I find that the latter turns out to be the case: the average song stays for longer in the charts, and this is driven exclusively by the weeks after a song reaches its peak popularity. Specifically, all of the increase happens when a song is in a rank in the bottom two thirds of the chart. This means that the songs seeing greater airplay increases are not the most popular ones, and these are potentially kept in the air after the point of satiation. Furthermore, I find some evidence for an increase in entry of new Canadian artists. And even though the quota increased the exposure some Canadian artists received, it seems that this increase was not matched with a proportional reception in US charts.

In order to argue whether a quota can be rationalized as an instrument to address consumption externalities or not, we must determine whether the valuation of music does exhibit dynamic complementarities. Exposure and familiarity dynamics are well studied in psychology and marketing. The central question in this literature is how repetition of a stimulus changes its subsequent evaluation, an effect labelled the "mere exposure effect" (Bornstein, 1989; Zajonc, 1968). The existence of this effect is well established for musical stimuli: exposure initially increases the valuation of a musical track, even when listeners retain no memory of what they heard, across different levels of musical complexity, and familiarity predicts song choice at least as well as stated preferences do (Madison and Schiölde, 2017; Szpunar et al., 2004; Ward et al., 2014). This notion is not new to economics. The canonical consumption or addiction capital framework developed by Becker and Murphy (1988) and Becker et al. (1991) laid the groundwork for modelling dynamically dependent preferences. This framework is particularly useful for studying cultural goods, where familiarity with niche-specific language and context can be important for valuation. Brito and Barros (2005) provides a notable exposition of how these dynamics can be modelled, while applications to theatre can be found in Lévy-Garboua and Montmarquette (1996) and Throsby et al. (2024).

The effect of exposure, however, need not remain positive as familiarity accumulates. It has been conjectured and documented that familiarity can have a non-monotonic effect on valuation, a phenomenon referred to as hedonic decline (Galak and Redden, 2018). Even in the mere exposure effect literature, the spacing of exposure and its duration moderate the effect size (Bornstein, 1989), in line with non-monotonicity. Such satiation with respect to past familiarity has also been documented in music (Chmiel and Schubert, 2017; Montoya et al., 2017). Specifically, Sguerra et al. (2022) analyzes music-streaming listening patterns among a sample of users and finds that the probability of listening to a track again is single-peaked with respect to the number of previous plays. This non-monotonicity has subsequently been incorporated into

two recent models of cultural-good consumption. Knight (2026) analyzes the problem of a monopolist curator allocating exposure between two goods when current demand for each good is single-peaked on past exposure. This non-monotonicity forces the curator to rotate exposure between the two goods in the optimal schedule even when one of them is strictly preferred in pure attribute terms. Lévy-Garboua et al. (2025) estimates experimentally a model of taste learning in which participants rate music tracks, and finds that the best-performing model formulation includes a satiation point for familiarity. Familiarity, then, is a documented and quantitatively important input into the valuation of music, with its effect turning from positive to negative as past exposure accumulates. I extend these findings empirically by showing that music-chart trajectories in the US and Canada exhibit a clear single-peaked relationship across the charting-spell distribution, as shown in Figure 2. This is only possible in a world where satiation exists and induces rotation and exploration of new hits.

A content quota differs from conventional forms of industrial policy because the regulator does not directly determine which producers receive support. It constrains the aggregate composition of the menu from which an intermediary chooses. The incidence of such policies can therefore depend on the intermediary's response. This has been documented in several settings. Retailers can adjust markups in response to tariffs, changing the extent to which trade policy reaches consumers (Cavallo et al., 2021; Cole and Eckel, 2018); firms subject to rules of origin can substitute toward qualifying inputs without expanding total output (Conconi et al., 2018); and fuel retailers can capture part of policy-induced subsidies through markups, with passthrough depending on market structure (Lade and Bushnell, 2019).

In each case, the regulated firm mediates the incidence of the policy. In the radio market, this mediation takes the form of selecting which Canadian works receive the mandated exposure. The policy therefore does not simply increase demand for Canadian music; it changes the constraint under which broadcasters choose their playlists.

The identity of the intermediary matters because broadcasters have strategic incentives of their own. Broadcasting markets are characterized by regulated entry and substantial scope for imitation across stations. Theoretical work has shown that competition can reduce product variety when firms strategically respond to one another's programming choices (Anderson and Coate, 2005; Spence and Owen, 1977; Steiner, 1952) and reduced form estimates point in this direction as well (Berry and Waldfogel, 2001; Berry et al., 2016; Berry and Waldfogel, 1999). Dynamic models of broadcasters have studied ad placement coordination and differentiation choices and the frictions in changing a format placement (Sweeting, 2009, 2013), while more general dynamic models have studied the incentives to experiment and learn about demand under competition (Aridor et al., 2025). These incentives can discourage exploration: a broadcaster that develops a successful new act may create value that competitors can appropriate by copying its programming. When exposure also builds consumption capital that is available to competing broadcasters, the private return to discovering and cultivating new acts is further reduced.

A content quota therefore changes the constraint on a strategic intermediary rather than simply shifting demand toward domestic producers. I use the 1999 reform to examine how broadcasters respond when the amount of Canadian content they must carry increases, contributing to both the industrial policy literature with intermediaries and the IO literature on broadcasting. I provide evidence that the behavior of the broadcasting

intermediaries is crucial for the effectiveness of the quota in providing exposure to Canadian artists.

Finally, this paper contributes to the literature on cultural quotas. Existing theoretical work studies how quotas alter the equilibrium allocation of cultural production and consumption. Richardson (2006), for example, models quotas as a distortion to the efficient allocation of broadcast content, while Richardson and Wilkie (2015) incorporates the responses of artists, record labels, broadcasters, and consumers. Empirical work finds that quotas can increase local content and affect when such content is broadcast (Filimon et al., 2011; Kyle and Niu, 2017), and the regulator’s own music-use study found Canadian selections concentrated in evenings, weekends and overnight hours (CRTC, 1998b).

This literature largely treats the quantity or composition of domestic content as the relevant outcome. I instead examine how mandated exposure is allocated across works and artists, and whether the additional exposure is broadly distributed or concentrated among commercially established acts. I additionally explore the export potential of the additional exposure by analyzing how well music that charts in Canada performs in the US. This distinction is central when exposure itself affects subsequent demand: a quota may increase the amount of Canadian music on the air without substantially broadening the set of Canadian music that reaches consumers. And, if there is no additional international penetration, then the extra airplay allocated to Canadian music would not be translating into better selection of artists but instead a reshuffling of the existing types of artists.

The rest of the paper is organized as follows. Section 2 describes the market structure of recording and broadcasting, and it describes the Canadian institutional setting. Section 3 describes the data sources and collection, and the construction of the main samples. Brief descriptive statistics and some empirical regularities are presented as well. Section 4 explains the main empirical exercises to be performed to measure the behavioral response of broadcasters and artists to the quota expansion, while Section 5 presents the main results. Section 6 concludes and discusses future work and potential policy alternatives.

2 Setting

2.1 Recording and Broadcasting Framework

This paper is set in the context of the music recording and broadcasting markets. These are both parts of a long vertical chain of suppliers. The starting block is the artist who creates a musical work. Typically, record labels provide financing and production support so that artists can get a market-ready version of their work (Caves, 2003). This step is analogous to venture capital being willing to finance a promising startup in exchange for a slice of future profits. The original way in which labels earned their revenue is from distributing the physical copies of the records they helped produce, paying a royalty back to the credited artists. This changed once digital formats became mainstream. An account of this evolution of the industry is given by Waldfogel (2014) and Waldfogel (2017). In this sense, the revenues of labels depend on the demand for each physical or digital copy of their portfolio of works net of distribution, production, and marketing costs. Artists on the other hand earn revenues proportional to record sales plus any performance fees, also net of production effort and costs.

The broadcasting market is linked through the need to reveal unobserved qualities of music to listeners. Radio stations earn revenues from selling advertising slots which are priced proportionally to the audience

volume, meaning their revenues are proportional to engagement. The demand each station faces is going to depend on the mix of music played, the quality of the non-music program (talk shows, news, etc.), and the willingness to change the station when either a song or program is not to the listener's liking³ At the same time, artists and labels need airplay because this gives them exposure to listeners either for them to learn how well they value the music and consider them in their choice set, or to accumulate consumption capital which would increase future demand. A change in broadcasting demand for music can, therefore, affect the short-run choices of existing artists and induce the entry of new artists, all mediated by the behavior of record labels who fund them.

Historically, labels would promote artists by signalling to radio programming directors which artists they are promoting, sometimes compensating them for it. This helps the broadcaster as it has more information about the quality of the music than no signal at all. Nonetheless, this signal is quite noisy: it has been documented that past success is not a predictor of future success, and a high share of records being promoted lose money for the label (Caves, 2003; Waldfogel, 2014). Therefore, the broadcaster has a reason to play music from artists backed by a record label, but it is still facing a substantial risk of playing a flop. In addition to this, the label can have some interest in overselling their product to recover production costs which can downplay the quality of the signal.

Competition between artists is a complex process as the choice of product placement is carried out on a sparse space where consumers have low awareness of the available options, meaning potentially similar products have very different demand based on how much each product is known. As well, placement choice is limited by copyright laws, meaning that an artist can only produce something so similar to the available catalogue. Broadcaster competition, on the other hand, has been studied systematically. The first element of this is the Steiner (1952) incentives to copy a competitor's program. This has been documented by Berry and Waldfogel (2001) and Berry et al. (2016), where competition decreases product variety. The incentive here follows from the fact that knowledge or exposure of an artist is "owned" by consumers. This means that in a world with industry charts a station that plays a song with untested demand takes a risk of losing engagement while a station that plays the same song the next period will know if it has a high or low demand and avoid the risk entirely. This means the station that takes the risk cannot appropriate the returns of finding a valuable song.

2.2 Canadian Institutions

The broadcasting and music industries in Canada have been subject to a varying array of regulations. For broadcasting this arises from the unique features of the market, where the regulator owns the wave space within which radio stations operate. For the music industry as a whole this arises from concerns over cultural sovereignty given the proximity of the United States. This section outlines the main regulatory features of these industries. Appendix C expands on these topics.

³The switching rate has been estimated at varying rates, with older estimates ranging up to 32%, while more recent estimates give a rate varying in the number of ads, with 0% for 1-ad breaks, increasing up to 7% and then decreasing to 4% (Michelon et al., 2024).

2.2.1 Content Regulations

The main content regulation in Canada that this paper is concerned with started in 1971, when the Canadian Radio-television and Telecommunications Commission (CRTC) introduced the first Canadian content (CanCon) requirements for commercial radio in AM frequencies. The original quota was a floor of 30% of weekly airplay that qualified as Canadian content. This requirement was later expanded to include FM frequencies in 1991. Before 1991, FM frequencies were subject to a station specific quota commitment, ranging from 10% to 30%. Qualifying as Canadian content required a stance on the criteria to classify a record as Canadian or not. For this purpose the CRTC established the MAPL system. A musical work qualifies as Canadian if it meets at least two of the following four criteria:

- (M) music is composed entirely by a Canadian
- (A) music is performed, or lyrics are performed, primarily by a Canadian
- (P) performance is recorded in Canada
- (L) lyrics are written entirely by a Canadian

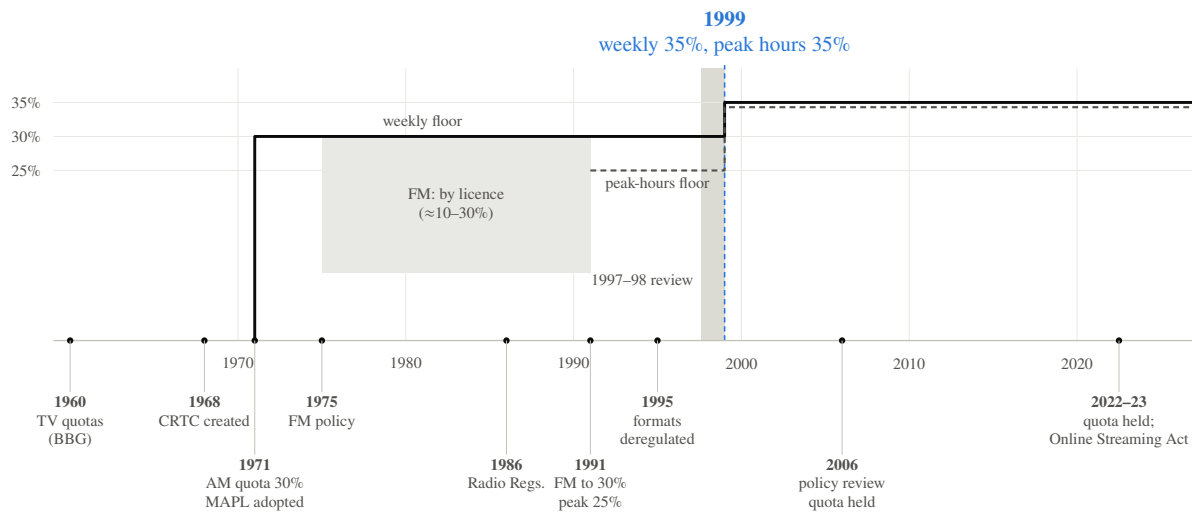
This system was later amended in 1991, where a record also qualifies as CanCon if both the lyrics and music credit 50% or more Canadian authorship (Regulations Amending the Radio Regulations, 1986, 1993). Note that the assignment is a function of characteristics of the work, meaning two works within a same album can have different CanCon status depending on who exactly contributed to its production and where exactly it was recorded.

The policy variation I will focus on in this paper is the content quota expansion implemented in January 1999. This was the product of the 1998 CRTC review of broadcasting regulations. The review was announced in August 1997, including public hearings in December 1997. The final report that contained the specific regulatory changes to be implemented was released in April 1998. This is the most recent change to CanCon requirements, which makes it the change with better catalogue coverage in the data sources I use. Analyzing prior requirement changes is a task for future work. The main changes introduced to programming choices by this review were:

1. The weekly quota was increased from 30% to 35% of weekly airplay.
2. Scheduling constraints were tightened from an expected floor of 25% of Canadian content during weekdays from 6AM to 7PM, to an absolute minimum of 35% between 6AM and 6PM. This made the requirement uniform during weekdays, equating the requirement between day time and night time.
3. To count towards the quota, a record must be played in its entirety, and not just a portion of it.

The nominal increase of weekly airplay, therefore, was bundled with a substantial tightening of how the quota was to be met, with an effective increase of 10pp during weekdays, implying a 40% increase on peak hours (CRTC, 1998b). Note, there is no floor during weekends, meaning that a station can in principle play only Canadian tracks during weekend nights and it would be in compliance with the regulations. Figure 1 summarizes the complete evolution of content regulations in Canada.

Figure 1: Canadian content regulation for commercial radio, 1960–2026



Notes: Minimum share of category 2 (popular music) selections that must be Canadian. Solid: the floor measured over the broadcast week (AM only until 1991, all commercial radio after). Dashed: the floor measured over weekday peak hours, 6 a.m.–7 p.m. from 1991 and 6 a.m.–6 p.m. from 1999; the two floors are equal from 1999 and the dashed line is drawn fractionally low so that both remain visible. Light band 1975–1991: FM obligations were set station by station, shown as an approximate range. Vertical band: the 1997–98 review.

2.2.2 Broadcasting Industry

The broadcasting industry in Canada is also regulated by the CRTC. The granular object being regulated are radio stations that live on a given frequency on either AM or FM bands. Entry is not free. To operate a radio station, a firm must apply for a license to the CRTC, and this is awarded if the applicant meets the requirements at the time. Note that these are not auctioned off, and there is no cost to apply except for making a case that the station does comply with the requirements. Because of this, the CRTC determines the number of participants in each market. In this sense, the profitability standards applied to the stations looking to enter cap the amount of participants indirectly. Ownership structure is explicitly regulated: the CRTC limited the maximum number of stations per owner to 1 AM station and 1 FM station per firm. Alongside with the quota expansion in 1999, the 1998 report extended the ownership limit to up to 3 stations per owner for markets with fewer than 8 stations, 2 per band, and in larger markets up to 2 for AM and 2 for FM. A market was defined for this purpose by the CRTC as the minimum between the area the station can serve and an audience measure that usually corresponds to a StatsCan Census Metropolitan Area (Radio Regulations, 1986).

In addition to regulating entry, the sale of a station has to be approved by the CRTC. As well, the buyer has to transfer part of the value of the sale as a condition for approval to third parties within the music industry. The amount transferred before 1999 was around 10% of the sale price but in practice it was negotiated case by case when the transaction was submitted for approval to the CRTC. After the 1998 review, the minimum was set at 6%, with 3pp going to the Radio Starmaker fund, 2pp to the FACTOR or MUSICACTION funds, and 1pp to the initiative of the buyer's choice (CRTC, 1998b, 2024a; Radio Starmaker Fund, 2024).

I have not analyzed how the ownership structure of stations responded to the loosening of ownership limits. It is likely this interacted with how broadcasters responded to the quota expansion. At the same time,

due to increased entry and erosion of retail sales, labels were also experiencing a substantial change in market structure, summarized in a consistent decline of revenues from the 90s to the launch of streaming services in the 2010s (Waldfogel, 2014).

3 Data

For this project I use a combination of music catalogues and music charts to track product characteristics and their demand. I compiled these datasets from publicly available sources. An overview of the main features of each of these sources plus the main data choices are also presented in this section. Further details can be found in Appendix D.

3.1 Music Catalogues

I track artists and their work using two broad crowdsourced music databases: MusicBrainz and Discogs.

3.1.1 MusicBrainz

MusicBrainz is a product of the MetaBrainz Foundation data ecosystem. Entries are provided by volunteer editors following a structured relational database (MusicBrainz, 2026b). The database is organized around works, which track the abstract musical composition. A work can be recorded several times in different places and by different artists. Each recording, as well, can be released in different formats and in different places. Works and recordings track artists who contributed to them, such as who was the music composer and who was the lyricist. These artists can be part of bands or groups or individual people (MusicBrainz, 2026e). Because of the voluntary nature of data entry, the coverage of MusicBrainz is not uniform across artists and works. In general, coverage will be skewed towards the interest distribution of the volunteer editors. Fortunately, this project is mainly concerned with tracking the commercial success of artists, for which coverage is much higher. The remaining limitation is an underrepresentation of smaller acts and niche genres, especially in older vintages. In the end, the true rate of missed artists and works is unobservable.

3.1.2 Discogs

Discogs, on the other hand, is a platform that mainly serves as a marketplace and Wiki for physical music releases, such as vinyl records, CDs, and cassettes, for music collectors (Discogs, 2026a). This naturally skews the coverage towards physical releases, and particularly those that have resale value. At the same time, the physical release focus guarantees better data quality for artist credits and release information, as each sale is matched to the existing records in the database and a seller can contribute a new entry if their item is not currently listed, as well as a collector who owns an item and it has no match in the data. Entries are peer-reviewed by the community and are usually revised. Users who are poorly rated face submission restrictions to keep a standard of quality of the database (Discogs, 2026b). The downside is that there is no explicit relational structure which makes it harder to track the same work across different releases, even by different artists. For sale purposes, Discogs includes a structured genre taxonomy with 15 broad categories

that then get a nested style tag for each posted release. This gives a structured categorization for different music types. A link between the two catalogues exists, with MusicBrainz having a field for Discogs IDs for a substantial share of its entries. I use this link to fetch the genre categorization from Discogs and to fill in missing credits information.

3.2 Music Charts

To measure demand for music, I use music charts. The ideal data sources would be the actual airplay data from individual radio stations and retail sales, but these are not publicly available. To address this, I compiled from publicly available music charts three different datasets that track the popularity of music in Canada, the US and the UK.

3.2.1 RPM

RPM was a Canadian music industry publication with weekly issues that ran from 1964 to 2000. It was the only publication to compile the music charts in Canada for most of its history. Some competitors emerged in the 80s and in the 90s, but it is the only one with presence around all CanCon changes. After its closure, Billboard took over this role in 2007. The charts were compiled in a rather artisanal way, with the magazine editor collecting data individually from radio station playlists, reports from labels, and reports from retail stores. The methodology changed over time, consolidating after 1990 to pure airplay charts for individual songs, and a combination of airplay and sales for albums. In 1997 the albums chart was powered by Nielsen SoundScan, but there is no evidence any other year used this method. The charts were published in the magazine, and are now available in digital format through Library and Archives Canada (RPM's Pop Charted Songs, 2018).

I extracted for the years 1994 to 2000 all the available charts using Google's LLM batch API. For this I first hand coded for each issue the page in which each type of chart appeared, and then used the API to extract from these chart-containing-pages the chart entries into structured tables. Table D4 summarizes the extraction process. I visually audited the data extracted and found that the extraction was accurate for text. The chart entry position numbers were deterministic as the LLM was told that they were extracting a ranked list of varying lengths. Any error in extraction, therefore, is concentrated on the artist and song title fields. This gives me a set of 6 weekly charts in Canada: RPM Top 100, RPM 100 Albums, RPM 100 Country, RPM Country Albums, RPM Dance, and RPM Adult Contemporary. There are other charts as well, but these do not have an overlapping coverage across the policy change window, and therefore are not used in any subsequent analysis. The charts selected for analysis are the ones that track individual songs' airplay, which are RPM Top 100, RPM 100 Country, RPM Dance, and RPM Adult Contemporary.

3.2.2 Billboard

Billboard is a US music industry publication that has been publishing weekly music charts since the 1950s. The charts are a composite measure of both airplay and sales, with a methodology that has changed over time. In 1991 Billboard shifted to a data-driven methodology accepting both the Nielsen Broadcasting Data

Systems (BDS) and Nielsen SoundScan data as measures of airplay and retail sales, respectively. BDS tracks airplay from a sample of radio stations, while SoundScan tracks sales from a sample of retail stores. In 1998 Billboard changed the criteria for inclusion in the Hot 100 chart to allow works with no physical release to be included. This was to address the labels’ concern that releasing a single might cannibalize album sales. At the same time, Billboard has a rule that after 20 weeks a song is removed from the chart if it is below position 50 (Billboard, 2016, 2018, 2026).

From the array of available charts, I selected 4 charts that match the styles in RPM for analysis. Album charts generally are excluded from analysis as they do not map to airplay directly, and the Hot 100 chart is also excluded as it faced a methodology change right in December 1998. The selected charts include Adult Contemporary, Alternative Airplay, Country Songs, and R&B/Hip-Hop Songs.

3.2.3 Official Charts

The Official Charts Company is a UK music industry organization that has been publishing weekly music charts since the 1950s. The charts are a sales measure. Measurement happens through participating stores over which sales are tracked in a sampling based approach similar to SoundScan. Because this approach does not capture airplay choices it is not included in most analyses.

Tables D4 and D5 of the data appendix summarize the main coverage and match rates of the chart datasets.

3.3 Dataset Construction

After collecting the music catalogue and chart data I construct a panel of artists and works that tracks artist and work characteristics through time. For the purposes of this research, I focus on primary credited artists. A credited artist is the entity that is tagged as having contributed to the creation of the musical work and receives some intellectual property rights. These can be either bands or individual people, and a single work can credit multiple artists. I focus on main credits, ignoring featured contributions or joint works for simplicity. Within this set of artists, I keep all those with at least one release between 1900 and 2010 in the MusicBrainz catalogue, excluding jazz and classical music because of their different industry dynamics. I then join each artist in this set to their set of released recordings. Recordings are deduplicated by title, stripping remasters and other accessory reissues, and within a tolerance of three seconds of duration. The objective is to capture the object that a radio DJ chooses to play, while retaining releases such as “live” or “remix” versions that may have sufficiently different attributes to constitute distinct products. I then match each release to its Discogs counterpart to obtain genre classifications and style labels.

I next match artists and works to the extracted charts data. I first match artist names and, within each artist’s catalogue of releases, match song titles, excluding ambiguous names such as “Various Artists” or “Unknown Artist”⁴. I leverage the nested structure of authorship to make the fuzzy string-matching procedure more efficient. Appendix D.1 provides further details on the matching process and the resulting dataset.

⁴I exclude from the matching the secondary credits such as featured artists or collaborations, as there is no systematic way to separate a & or / sign from part of an artist name (Earth, Wind & Fire or AC/DC, for instance).

MusicBrainz also provides the country of origin of each credited artist, which I use as a proxy for CanCon compliance, since the MAPL score is weakly increasing in Canadian origin.

The resulting dataset allows me to follow the charting performance of each matched work for each artist in each week and on each chart, while also observing the artist's country of origin and the cohort of the work's first release. This allows me to identify entry into recorded music and examine how newly released works perform in the Canadian and US charts.

3.4 Descriptive Statistics

Table 1 summarizes the main features of the constructed dataset. Panel A displays how well the linkage of all the data sources worked and what is the coverage of the main treatment variable: the country of origin. Across all the artists in the MusicBrainz catalogue some 65% of them present an origin country tag. All subsequent analysis has to rely on this subset of artists. Figure A1 in the appendix expands on this coverage. The figure shows two processes: first, the amount of artists covered by MusicBrainz increases over time and its increase accelerates strongly at the end of the 80s. This is a documented feature of the music industry, as told by Waldfoegel (2017), where both cheaper production and distribution technologies have allowed widespread entry into the industry. The second process is how this increase in artists is not matched by the coverage of the country tag, displayed in a decreasing share of each cohort possessing a country tag. A potential explanation for this is that despite an online catalogue being biased towards more recent artists, the bounded amount of volunteer editors cannot keep up with the pace of new artists releasing music, so some of the entries are not fully documented. At the same time, the entries with incomplete information are probably skewed towards less popular artists, which are less likely to either chart or have retail sales in their home country or abroad.

The country tag filter keeps 245,138 artists, of which 24.7% have released some music in the current year or the two years prior for 1994 to 1997. This smaller share is also driven by the growth of the catalogue, so the majority of the registered artists are from more recent cohorts. Within the active artists, as well, the linkage to Discogs is 73% of the artists having a Discogs entry to supplement additional metadata.

In line with the catalogue growth story, the average and median debut years are relatively recent, 1997 and 2001, and when we keep only the ones active close to the CanCon expansion in 1999 they drop to 1990 and 1994. Active artists have larger catalogues, and charting artists even more so. This is unsurprising as releases are a function of how much commercial potential an artist has. Regarding the charts coverage, we have that 94.9% and 98.5% of RPM and Billboard entries are matched to an artist in the selected song charts. At the same time this maps to around 75% of distinct artists and song titles matched in RPM and a somewhat higher coverage in Billboard. Overall, most chart entries can be assigned to an artist, plus 91% and 87% of these entries can be matched to a country tag, meaning within chart coverage is also relatively complete.

Table 2 splits the main descriptive statistics between Canadian and non-Canadian credited artists. This implicitly keeps only artists with a known country of origin. I also zoom into the active artists in the years prior to the expansion of CanCon as these will be the most likely artists to be competing for airplay.

The first pattern arising from the table is that Canadian artists are a small population relative to the rest of the catalogue and they have smaller and younger catalogues. With an overall population of 8,911 artists relative to 236,227 non-Canadian ones, and a 2.6 years later average debut year, Canadian artists show an

Table 1: Catalogue breadth and chart-match coverage

	(1) All artists	(2) Active 1994–97	(3) On RPM	(4) On Billboard
<i>Panel A. Artist coverage (share of the column's artists, %)</i>				
With a country tag	65.0	73.5	90.5	87.1
Present in Discogs	58.1	73.0	90.3	87.0
Active, 1994–97	21.9	100.0	77.2	39.2
Matched to RPM	0.7	2.6	100.0	8.7
Matched to Billboard	5.7	10.2	66.5	100.0
<i>Panel B. Catalogue per artist, mean (median)</i>				
Debut year	1996.8 (2001)	1990.4 (1994)	1989.4 (1993)	1990.9 (1995)
Releases	4.7 (1)	11.9 (3)	65.1 (14)	25.7 (6)
Release-tracks (raw)	49.5 (14)	136.7 (30)	754.5 (128)	292.6 (61)
Songs (deduplicated)	31.9 (13)	74.5 (26)	235.6 (79)	118.5 (42)
Years with a release	2.6 (1)	5.1 (3)	10.1 (7)	7.1 (4)
Years between releases	3.7 (2)	3.6 (2)	2.5 (2)	3.4 (2)
With ≥ 2 release years	44.1	66.5	85.0	76.9
<i>Panel C. Chart-side match coverage (share of chart entries, %)</i>				
Chart entries (slot-weeks)	—	—	94,359	88,010
Entry artist resolved	—	—	94.9	98.5
Entry artist and title resolved	—	—	79.0	87.6
Distinct chart artist names	—	—	4,905	2,520
resolved to a catalogue artist	—	—	74.6	95.8
Distinct chart titles	—	—	5,808	5,579
resolved to a catalogue song	—	—	75.8	85.2
<i>Artists</i>	377,323	82,571	2,798	21,484
<i>Artists dropped for want of a country tag</i>	132,185	21,908	266	2,766
<i>Songs (deduplicated)</i>	12,051,608	6,151,706	659,291	2,544,853

Notes: The unit is a credited artist in the merged MusicBrainz–Discogs catalogue. Classical, jazz and non-music releases are excluded, as are 17 MusicBrainz special-purpose entities such as *Various Artists*. Column (1) is the whole catalogue; (2) artists active in 1994–1997, meaning a deduplicated song first released 1992–1997; (3) and (4) artists resolving to at least one chart entry whose artist and title are both matched, or ambiguous scoring at least 85 of 100, over each source's full chart history: Billboard 1956–2026, RPM 1994–2000. The columns are not nested in one another. Country tag is MusicBrainz's country field for the credited artist, the paper's treatment variable. Present in Discogs is any Discogs-derived field on the artist: genre tags, a credit, or a release country. Panel A rows are shares of the column's artists. Panel B gives the mean with the median in parentheses; release-tracks count every occurrence on every release, songs count each once at its earliest release year after deduplication on title and duration, and years between releases is the first-to-last span over one fewer than the number of release years, defined for artists with at least two. Panel C is computed over chart entries from 10 January 1994 to 6 November 2000 on RPM's four song charts (Top 100, Country, Adult Contemporary, Dance) and Billboard's track charts excluding the Hot 100; a chart name or title counts as resolved if any of its entries does.

expectably younger catalogue in 1999, but also a lower count of releases and of distinct songs, despite having a similar time span between each release. This can be explained by the mere difference in age of the artists, as releases accumulate over active career time. However, among active artists the differences persist despite a narrower gap in debut year, suggesting either a composition difference or a recording industry difference.

The second pattern is that Canadian artists are more likely to chart in RPM by a large margin, and despite having a lower overall entry rate into Billboard, they tend to chart longer and higher than non-Canadians⁵.

⁵Only 32 Canadian artists appear in the selected Billboard charts between 1994 and 2000, so this trend is based on a small sample.

Table 2: Canadian and non-Canadian artists

	Full sample			Active 1994–97		
	Canadian	Non-Canadian	<i>N</i>	Canadian	Non-Canadian	<i>N</i>
<i>Panel A. Catalogue and composition</i>						
Debut year	1998.5	1995.9	8,911 / 236,227	1990.7	1989.1	1,873 / 58,790
Catalogue age at 1999 (years)	0.55	3.06	8,911 / 236,227	8.29	9.85	1,873 / 58,790
Releases	4.71	6.36	8,911 / 236,227	11.23	15.50	1,873 / 58,790
Release-tracks (raw)	50.50	68.04	8,911 / 236,227	131.06	179.13	1,873 / 58,790
Songs (deduplicated)	31.42	41.38	8,911 / 236,227	67.48	94.71	1,873 / 58,790
Years with a release	2.62	3.16	8,911 / 236,227	5.14	6.33	1,873 / 58,790
Years between releases	3.55	3.75	4,372 / 126,311	3.60	3.57	1,333 / 45,194
With ≥ 2 release years	49.1	53.5	8,911 / 236,227	71.2	76.9	1,873 / 58,790
<i>Panel B. Chart presence, 1994–2000</i>						
On RPM, 1994–2000	2.9	0.4	8,911 / 236,227	11.9	1.5	1,873 / 58,790
On Billboard, 1994–2000	0.4	0.6	8,911 / 236,227	1.3	2.1	1,873 / 58,790
RPM share of slot-weeks (%)	0.080	0.056	258 / 1,005	0.085	0.061	223 / 875
RPM distinct works	4.05	3.25	258 / 1,005	4.34	3.51	223 / 875
RPM best position	0.750	0.738	258 / 1,005	0.748	0.746	223 / 875
Billboard share of slot-weeks (%)	0.097	0.056	32 / 1,482	0.117	0.061	25 / 1,255
Billboard distinct works	4.12	3.17	32 / 1,482	4.76	3.44	25 / 1,255
Billboard best position	0.652	0.612	32 / 1,482	0.724	0.623	25 / 1,255
<i>Artists</i>	8,911	236,227		1,873	58,790	

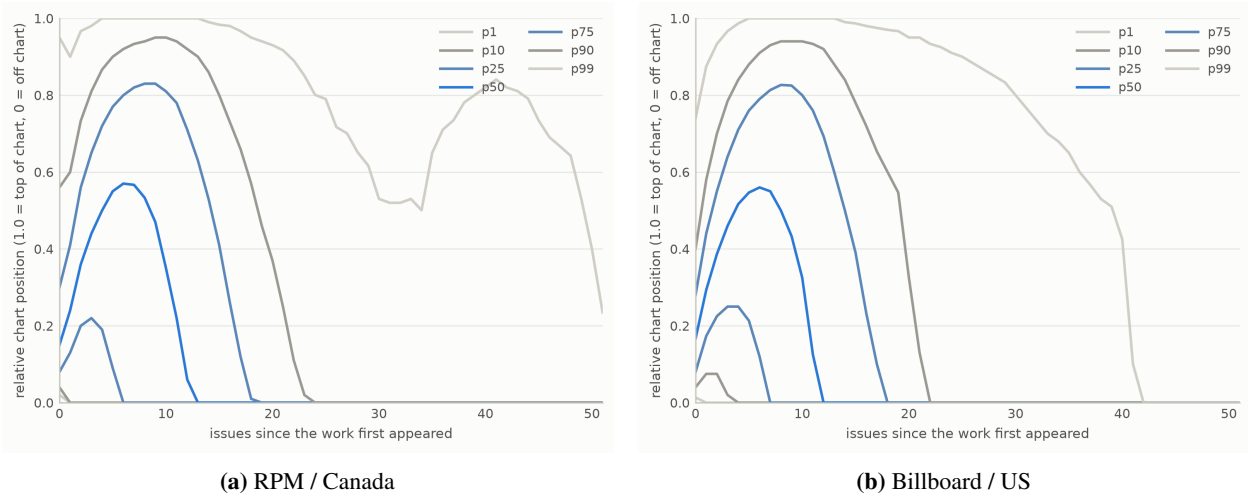
Notes: The unit is a credited artist. The full sample is every artist carrying a country tag, 8,911 Canadian and 236,227 non-Canadian. The active sample restricts that to artists active in 1994–1997, meaning a deduplicated song first released in that year or the two before it, so at least one first released 1992–1997: 1,873 and 58,790. Non-Canadian is every other country tag. MusicBrainz special-purpose entities are excluded. Percentage rows are shares of the column’s artists, except share of slot-weeks: the chart $\times$ issue slots the artist held over every slot that source published in the window, counting unresolved entries (94,355 on RPM, 88,010 on Billboard), in percent. “Years between releases” is the span from first to last release year divided by one fewer than the number of release years, and is defined only for artists with at least two. The indented rows of Panel B are conditional on the artist appearing on that source at all, so their denominator is the share reported in the row above and not the column’s full sample; a non-appearing artist is missing there rather than zero, which is why *N* is given per row rather than per column. Chart presence is measured on this paper’s own chart universe over 10 January 1994 to 6 November 2000: RPM’s four continuously-published song charts (Top 100, Country, Adult Contemporary, Dance) and Billboard’s track charts excluding the Hot 100. Best position is the artist’s highest *relative* position, $(\text{depth} - \text{rank} + 1) / \text{depth}$, where 1 is the top slot of a chart, $1 / \text{depth}$ its bottom slot and 0 is off the chart altogether. Depth varies across charts and within the window: RPM’s Top 100 has a hundred slots and Billboard’s Adult Contemporary thirty. Depth is the page’s real slot count, taken from every row of that chart-week including the ones whose artist did not resolve.

This is probably due to selection, as only the most commercially appealing artists break into the US market. In RPM, on the other hand, there are no differences in the highest chart position despite occupying a higher share of the available slots each week and having an overall higher probability of charting. This higher probability could reflect the extent of CanCon in airplay programming, allowing a bigger share of Canadian artists to chart, or it could be reflecting home-bias in the Canadian broadcasting market.

3.5 Charting Regularities

In addition to catalogue level statistics, it is worthwhile diving into some of the regularities that aggregate airplay measures show. I motivated a rationale for government intervention in the broadcasting industry based on the possibility of a consumption capital dynamic complementarity in consumer tastes. It is straightforward to picture that learning is a feature of this market as the amount of products available makes it impossible

Figure 2: Chart trajectory of tracks



Notes: Percentiles of relative chart position at each issue since a work first appears, pooled over every chart of the market and over its whole matched span, with no split by origin, period or chart: Billboard / US 1958–2009, 2,674 issues over 4 charts; RPM / Canada 1994–2000, 324 issues over 4 charts. The UK Official Charts, which rank sales rather than airplay, and Billboard’s Hot 100, whose methodology changed in December 1998, are excluded. Relative position is $(\text{depth} - \text{rank} + 1) / \text{depth}$, so 1.0 is the top slot of a chart and charts of different depth are commensurable. The cohort is fixed: every work is followed the same 52 issues from its first appearance, and an issue off the chart is encoded as 0, strictly below the lowest available slot, so the set of works behind every percentile is the same at every k . Works whose 52 issues do not fit inside the market’s own calendar are not in the cohort. Cohort size: Billboard / US $N = 32,097$ works; RPM / Canada $N = 3,793$ works.

for most consumers to have a good notion of what is the choice set and how much they value each option. This justifies the signalling role of broadcasting and label advertising, which would help consumers navigate through the choice set. However, a prediction of pure learning is a relatively stable ranking of airplay: if consumers truly care only about pure unobserved quality, once they know which products are most valued they will always prefer the most valued products, which would translate into broadcasters playing the most popular songs for their respective demographics on repeat. An extreme version of this argument suggests that each station should play exactly one song all the time, as this would maximize revenues for their demographics.

Real world experience and Figure 2 show this is not the case. The figure shows a clear diffusion of the popularity of a track as it climbs the charts, and a clear decline in popularity afterwards. Particularly, this is a property of the whole distribution of chart trajectories, as low, medium and high quantiles show the same single-peaked trajectory. This pattern is consistent with a process of consumption capital accumulation and later satiation. This has also been documented at the individual song-choice level, where Sguerra et al. (2022) measures a similar hump-shaped probability of playing a song as a function of how many times it has been played already. The increase in popularity can also be explained by some mimicking strategies between radio stations: as a song enters the chart, other stations realize it is a profitable choice and play it more often, which makes the song climb the charts higher and signal even more quality to other stations. This cannot explain the subsequent decline, nonetheless. Therefore, this is consistent with other prior estimates of consumption capital accumulation in music, and suggests that broadcasters rotate the hits being played across the board, with 90% of songs dropping out of charts after 25 weeks.

A dynamic complementarity with consumption capital also predicts that songs that leave the chart should

have a positive probability of returning that is increasing in the time spent off the chart: as the accumulated capital depreciates its level should decline below the level of satiation, allowing for a potential reentry. Figure A3 shows a small and decreasing probability of reentry, which is consistent with decaying consumption capital resetting below the satiation level, and at the same time both the decrease in promotion and the same decaying consumption capital making the song less appealing. A pure learning story predicts a zero probability of reentry, as the song's value and demand level is already known and it has been deemed not worthy of a chart slot. At most, this would allow for some reentry given composition changes in the audience or seasonal taste shocks, such as holiday songs getting popular during holidays, but not a systematic positive reentry probability.

A further feature of this market is that tastes evolve over time. Figure 3 plots the relative amount of slots of both RPM and Billboard charts occupied by some popular music styles. The first pattern to notice is that chart composition is not stable over time, with Disco surging in the late 70s, Soul declining, and Alternative Rock rising in the 90s. The second feature is that this evolution is relatively slow. The rise of Alternative Rock in Billboard took between 5 and 10 years to consolidate. The whole Disco run took 20 years from early rise to late decline. This is confirmed in the evolution in RPM during the 1994–2000 window: there is no stark change in the relative rank across styles.

Therefore, the constructed dataset allows me to track aggregate airplay measures and entry patterns across artists of different origin with a relatively good coverage, but airplay has a strong local dynamic component that has to be accounted for. This same dynamic component can also be an outcome of interest: instead of just measuring the probability of charting, I can look into the length of charting spells.

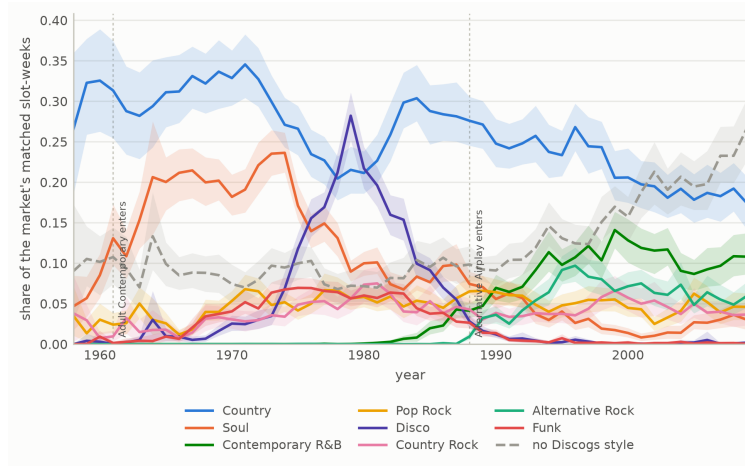
The ideal direction for future work is to obtain airplay logs from radio stations. Charts are an order statistic measure capturing only the right tail of the airplay distribution and in an aggregate manner. Airplay logs would allow a much finer analysis and a better understanding of the broadcasting choices for non-hit tracks.

4 Empirical Strategy

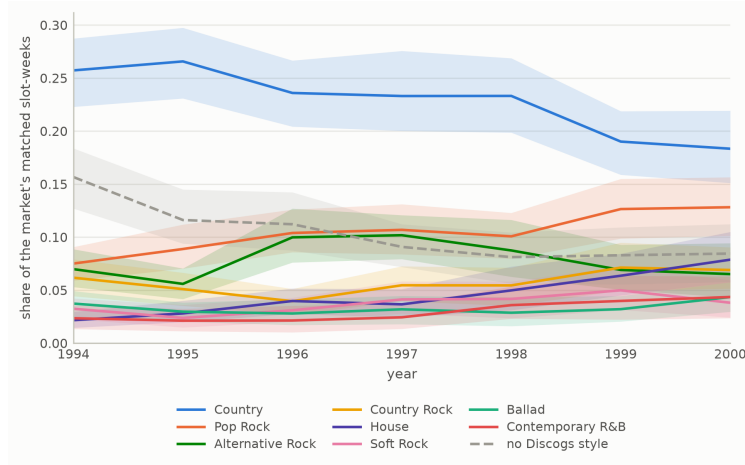
There are several questions we need to answer to understand what are the behavioral responses to a content quota. The first is whether an active artist is more likely to receive any of the mandated airplay. A natural expectation from a content quota is that the affected group will receive more airplay in general. However, if the broadcaster can substitute within the affected group, the sign of the effect is not clear a priori: the amount of airplay is fixed meaning an increase in concentration would cause an average decrease in airplay per artist. The second question that arises is how does airplay in itself change. The distribution of who gets airplay probably changes, but also it is likely that either the dynamics or the characteristics of airplay also change: maybe now broadcasters play less music overall to avoid risky artists, or maybe they bring back older songs from established acts to comply with the quota in a safe manner.

Finally, as broadcasting changes the information layout consumers have, the quota also affects the entry incentives of new artists and how they coordinate with labels. A natural conjecture here is that the quota raises expected revenues and therefore, we should see an increase in entry flows of new artists.

Figure 3: Discogs styles' share of matched airtime, by year



(a) Billboard / US



(b) RPM / Canada

Notes: Each line is one Discogs style's share of the market's matched slot-weeks in that year. A slot-week is one work on one chart of the market in one issue. The styles drawn are the 8 largest by total matched airtime in that market over its whole span, chosen within the market, so the two panels need not carry the same styles: RPM / Canada Country, Pop Rock, Alternative Rock, Country Rock, House, Soft Rock, Ballad, Contemporary R&B; Billboard / US Country, Soul, Contemporary R&B, Pop Rock, Disco, Country Rock, Alternative Rock, Funk. A work carries several styles and its slot-weeks are split equally across them, so the shares of all styles sum to one; the residual of the styles drawn is not plotted. The dashed grey line is the share held by works to which Discogs attaches no style, kept as its own level rather than dropped. Styles are the release's own Discogs labels, joined to chart entries through the matched artist and title. The denominator is matched slot-weeks and not published slots, so a level is conditional on the match: Billboard / US 6.8% to 26.6% of matched slot-weeks carry no style; RPM / Canada 8.1% to 15.7%. Bands are 95% intervals from resampling works with replacement within market $\times$ year, 200 draws, the work being the unit a slot-week is not independent within. Dotted rules mark the first matched issue of each chart, which is where the pooled denominator gains that chart: Billboard / US Adult Contemporary 1961, Alternative Airplay 1988. Years holding fewer than 200 matched slot-weeks are not drawn. Sample: Billboard / US 52 years, 453,206 matched slot-weeks; RPM / Canada 7 years, 72,847 matched slot-weeks.

4.1 Entry into charting

To measure whether the quota mandated airplay was allocated to Canadian vs foreign artists, we can leverage the fact that eligibility to the quota is weakly increasing on the credited artist having a Canadian origin⁶. This is desirable because country of origin is a fixed attribute of credited artists and therefore cannot be manipulated or respond to the quota. As well, the other criteria for eligibility can cause behavioral responses: artists can choose to produce records with more Canadian lyricists or composers, or to move production to Canada. This would make the raw MAPL score endogenous to the quota. At the same time, country of origin is the best measured dimension of the score available in the data.

Therefore, if Canadian and foreign artists would follow similar trends of charting probability in the absence of the quota, the difference in the probability of charting between Canadian and foreign artists before and after 1999 can be interpreted as the causal effect of the quota increase⁷.

The estimation has to account for three issues. First, the CanCon expansion was announced in August 1997, holding hearings through December 1997, and the full specification of the review was released in April 1998. This leaves enough time for anticipatory responses. For this reason I fix the reference period to 1997Q2. The second issue is that the probability of charting is a function of the life cycle of an artist. For this I include cohort-period fixed effects, implicitly controlling for age for each artist. Third, the probability of charting is a ratio of successful counts relative to the total count of artists in the sample. I fix the set of artists to the artists that released at least 1 track from 1992 to 1997 to account for this. As is shown in Figure A1, cohorts are increasing in size over time, which could still bias the charting probability difference downward if Canada was growing faster than the rest of the world as there would be a larger pool of competing Canadians. For this I also estimate a specification allowing for entry up to the end of the sample period, and another controlling for Canadian and non-Canadian artist pool growth trends.

The equation estimated is:

$$Y_{at} = \alpha_a + \delta_{tk(a)} + \sum_{e \neq 1997Q2} \theta_e (CA_a \times \mathbf{1}[e(t) = e]) + \varepsilon_{at} \quad (1)$$

Here Y_{at} is an indicator for an artist a of cohort k charting at least once in any of the 4 selected song charts of RPM in the quarter t or the sum of chart entries per week the artist appeared in for these charts in quarter t . α_a are artist fixed effects, $\delta_{tk(a)}$ are cohort-quarter fixed effects, and CA_a is an indicator for the credited artist a being Canadian. The cohort k is defined by the first year of a song release by the artist. The coefficients of interest are θ_e , which measure the difference in the probability of charting between Canadian and non-Canadian artists in quarter e relative to the reference quarter 1997Q2.

Note as well that the dynamics shown in Figure 2 make errors serially correlated across weeks. The choice of a quarter as the period aims to average out these while keeping enough periods to see time shocks. The ideal method would estimate a dynamic panel, but charting is a rare event so there is little power to estimate

⁶The specific rule is that 2 of the following conditions are satisfied: the main performer, that is the lead singer in most cases, is Canadian, the composer is Canadian, the lyricist is Canadian, or the performance took place in Canada. A credited artist having a Canadian origin tag satisfies the performer condition, and increases the score by 1.

⁷Inference is currently implemented as regular cluster robust, but the single treatment stratum and the finite amount of charts and slots suggest some finite inference methods should be considered.

this, plus the shape of the trajectory is both a known non-monotonic shape interacted with an artist/song fixed effect. Because these are rare events, most artists and songs chart only once.

4.2 Charting outcomes

Having measured the effect of the CanCon expansion on entry to charting, we can also characterize what is the response within charting artists and songs. For this I built a pooled chart-run dataset that tracks all charted songs in all the selected RPM song charts. I also built a charted-artists panel following quarterly each artist that charted in an RPM song chart.

Here I estimate a DiD model as well, only the sample is not a panel and composition varies as each observation is a chart-run, which is a unique event. The equation estimated is:

$$y_s = \alpha_{a(s)} + \delta_{t(s)} + \gamma_{c(s)} + \sum_{e \neq 1997Q2} \theta_e (CA_{a(s)} \times \mathbf{1}[e(s) = e]) + \epsilon_s, \quad (2)$$

where y_s is the outcome of interest for a chart-run s , $\alpha_{a(s)}$ are artist fixed effects, $\delta_{t(s)}$ are quarter fixed effects, and $\gamma_{c(s)}$ are chart fixed effects. The outcomes of interest are the total length of a charting spell, the number of weeks from spell start to peak position, the number of weeks from peak position to spell end, peak position, song age at spell start, an indicator for whether the song is an artist's best hit or a back catalogue song, an indicator for whether the artist is in their debut year, and artist career age at the start of the spell.

Note that these outcomes are aiming at specific aspects of what a consumption capital driven rotation would predict for broadcasters when faced with a quota. The chart run length and maximum position are measuring changes to the shape of the chart trajectory shown in Figure 2. Song and artist age, and the best hit indicator aim to measure if the songs played are actual songs intended to be hits or are picks from the back catalogue of artists that happen to comply with the new quota, or whether the charting songs are from debutant artists.

I show both the pooled and dynamic DiD estimates. The dynamic estimates have the issue of charting dynamics being part of the error term, which increases the errors. The pooled effects, in this sense, average out these dynamics. As well, the choice of charting spells is intended to abstract away from these dynamics. However, if the quota is to affect the shape of these dynamics, the spell-level estimates will have embedded a reweighting of the sample that is also part of the response by broadcasters.

4.3 Charting trajectories

To decompose the effect of the quota expansion across the charting trajectory, we can leverage the known structure of a charting spell and compare within each week of a spell before and after the policy. This allows us to see exactly where in the spell did the quota expansion have an effect. For this I build a spell-age panel that tracks the relative position, and indicator for charting of each song.

The equation estimated is:

$$y_{sk} = \alpha_s + \delta_{t(s,k)} + \gamma_{c(s)k} + \lambda_{kP(s)} + \mu_{kCA_{a(s)}} + \sum_{k' \neq 0} \theta_{k'} (CA_{a(s)} \times P_s \times \mathbf{1}[k = k']) + \varepsilon_{sk}, \quad (3)$$

where y_{sk} is the relative chart position, or an indicator for charting, of song s in the k -th issue since its first appearance, $k = 0, \dots, 25$, and is zero when the song is off the charts. P_s is an indicator for song s first appearing on or after January 1999, α_s are song fixed effects, $\delta_{t(s,k)}$ are quarter fixed effects for the issue, $\gamma_{c(s)k}$ are chart $\times$ issue-since-entry fixed effects, and $\lambda_{kP(s)}$ and $\mu_{kCA_{a(s)}}$ are issue-since-entry fixed effects by entry period and by origin.

4.4 Entry into recording

A first attempt to measure how entry responded to the CanCon expansion in 1999 is to use a synthetic controls estimator in the style of Abadie et al. (2010). The key challenge here is that the whole country is treated and the outcome is an aggregate count of artists instead of a chart or artist outcome. Therefore, we can get a measure of how much Canadian artist count changed by constructing a synthetic counterfactual series by minimizing the distance between pre-treatment Canadian artist flows plus a set of covariates with a weighted average of a set of donor countries. The key assumption of this is that first, Canada did not experience any other shock besides the expansion in 1999. The second one is that the donor countries are both not affected by the Canadian policy in question and did not face their own common shocks.

The outcome of interest will be $y_{ct} = \log(n_{ct})$, where n_{ct} is the cumulative stock of artists releasing music in country c since 1950 taken from the MusicBrainz catalogue, and t is the calendar year. The outcome choice follows from the fact that the artist stock reflects not only the size of the recording industry, but also it proxies for population size, education, and economic development, which means that matching on the trend of the artist stock is implicitly also matching the trend of these other variables, providing a more robust counterfactual construction. Alternatively, constructing a synthetic control over the flow of new artists would fit the weights on not only the trend of the entertainment industry, but also on any temporal shocks occurring over time. This ignores, or differences away any persistent component that could also determine the entry flows, such as population size. Canada is the treated unit and the donor pool $\mathcal{D}$ is the 29 countries holding at least 1,000 artists in the panel and a non-zero debut count in every pre-treatment year. Donor weights are estimated by finding the minimum RMSE fit of the pre-treatment period over the donor pool weight simplex:

$$\hat{w} = \arg \min_{w \in \Delta^{|\mathcal{D}|}} \sum_{t=1975}^{1996} \left(y_{CA,t} - \sum_{c \in \mathcal{D}} w_c y_{ct} \right)^2, \quad \hat{\tau}_t = y_{CA,t} - \sum_{c \in \mathcal{D}} \hat{w}_c y_{ct}, \quad (4)$$

with Δ the unit simplex, so the counterfactual is a weighted average of observed countries. The weights are fitted on 1975–1996 only; 1997 and 1998 are excluded from the fit to allow for anticipatory responses. The effect $\hat{\tau}_t$ is computed over 1999–2010. Inference is a conformal p -value following the methodology of Chernozhukov et al. (2021). As well, a standard placebo test is performed by applying the same test to Canada in a year in which no CRTC policy change occurred, such as 1981, as in Abadie et al. (2015).

5 Results

I first report the results on the probability of charting for a given Canadian artist relative to a non-Canadian artist. Figure 4 shows that an artist that had released some records between 1992 and 1997, after controlling for cohort-period shocks and artist specific shocks, is less likely to chart in any selected RPM song chart after the quota expansion, with some anticipatory effects. The pooled effect is a reduction of 1.67 pp, relative to a pre treatment quarterly charting rate of 4.2% for Canadian artists, implying a 40% reduction. At the same time, I find no significant difference on the total number of chart entries occupied by Canadian artists, with a pooled point estimate of -0.07 and a pre-treatment mean of 0.437. This implies that the average charting Canadian artist has to chart over more weeks or it has more songs charting at the same time.

Panels (b) and (d) of the same figure show that the estimates are unchanged by allowing for entry of artists up to the year 2000 and the effect is similar, only slightly smaller, meaning the effect is not driven by competition with new artists⁸. Artists that had not released any records after 1992 chart at a rate of 0 for Canadian and non Canadian artists, which means they could not drive the effect either.

This is a large reduction. And it is also a counterintuitive result as in principle a quota should expand the access to airplay occupied by Canadian artists relative to non-Canadian ones. However, this is only the case if the quota is allocated homogeneously across all Canadian artists. As discussed previously, broadcasters are not choosing how to allocate airplay slots neutrally. Therefore, they are more than willing to substitute between Canadian artists, and if the quota is allocated mainly on a minority of artists while displacing the rest of Canadian artists, the average effect would be negative while total chart presence remains the same or increases. At the same time, a trend in the pre-period could be driving this. For instance, suppose the record label industry had been getting more coordinated in the promotion efforts so even before the policy fewer artists are getting promoted on every release. This would show up as a negative effect but mainly due to a process unrelated to the policy. To account for any such pre-period persistent process, Figure B1 estimates the same pooled effect controlling for a linear trend fitted in the pre-period for Canadian and non-Canadian artist pools. The results are much less precisely estimated as a Canada–time variable is more collinear with treatment, but the magnitude and sign of the effects are similar to the main estimates. This and the fact that the second panel shows almost the same results rules out entry or other artist pool changes as the main driver of the negative effect.

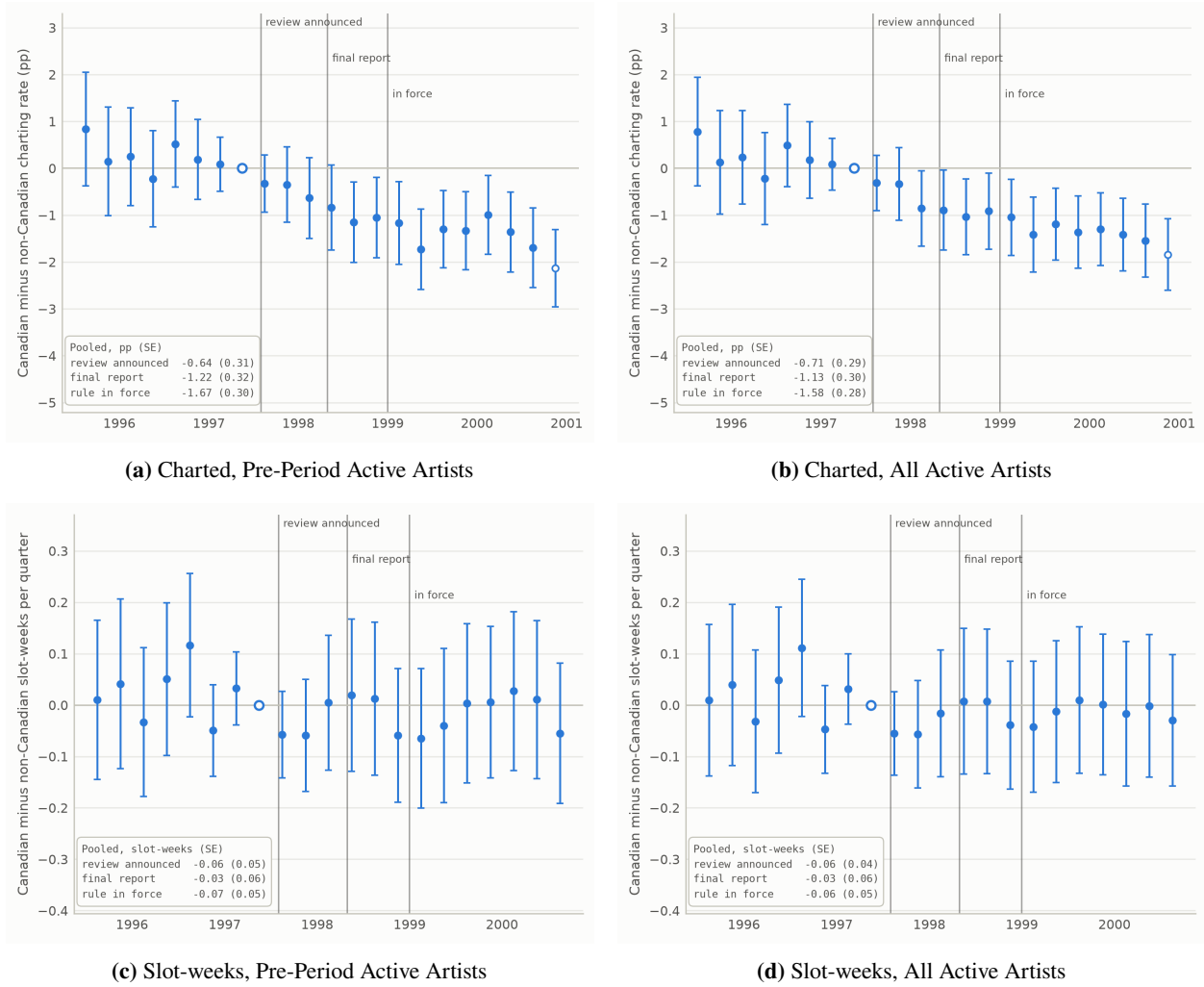
This extensive margin result does not say anything about how much each artist or song is being played, or if the typical chart run dynamics of a song has changed. To measure this, Figure 5 and Tables 3 and 4 show the results over songs and artists that charted at some point between January 1994 and November 2000. For these outcomes I estimate equation 2 over a pooled sample of uncensored chart spells⁹.

The first takeaway is that the average artist that charted at some point is not more likely to chart at a higher entry or peak position after the expansion of the quota. The results show, as well, a positive effect on the total run length, with this effect decomposed into an insignificant decrease in time to reach a peak position,

⁸Canadian artists who chart do so 14% of the time in their debut year, and 45% in their first 3 years, so charting is concentrated among entrants.

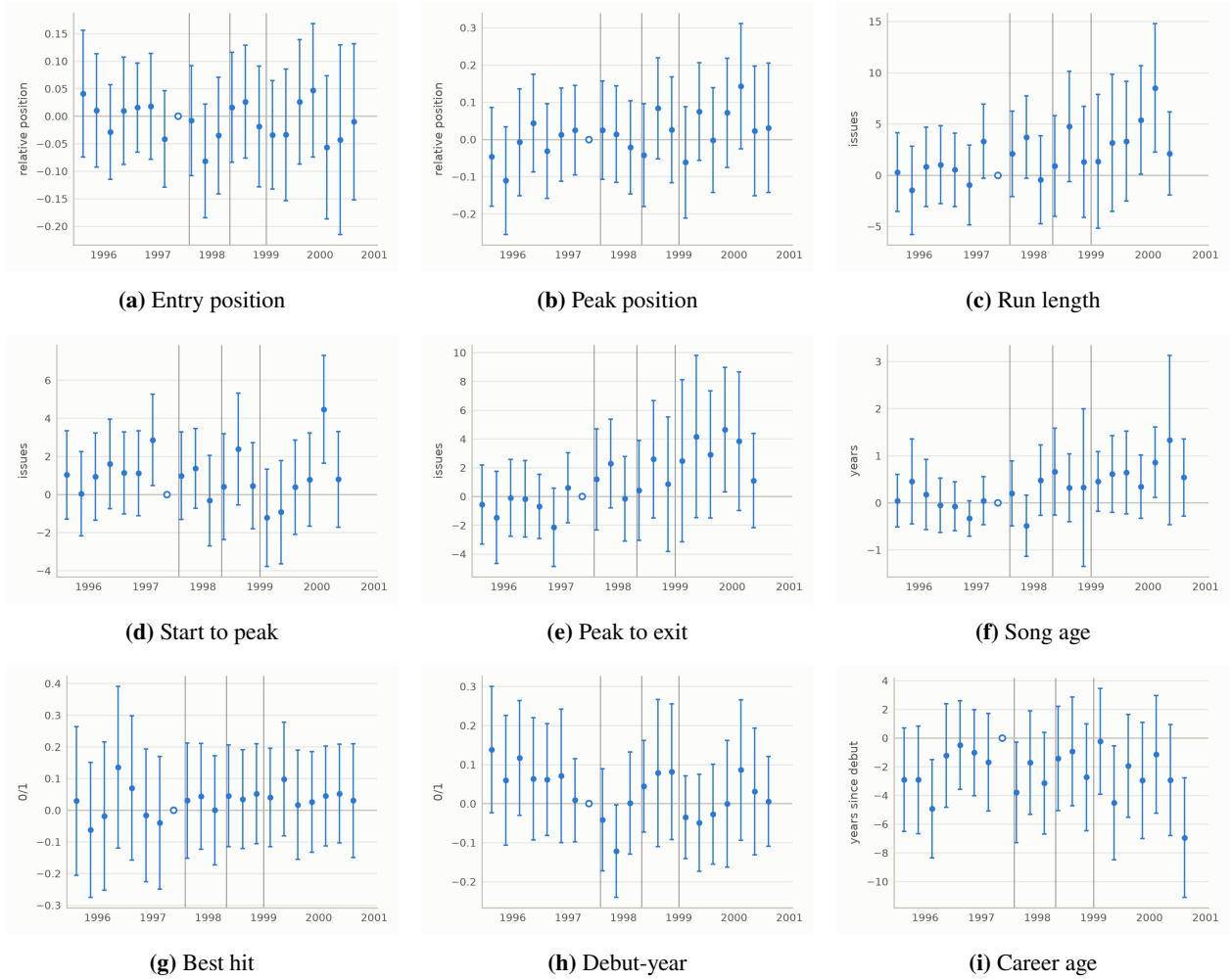
⁹Some spells have already started at the start of the charting period in 1994 making them left censored, while some spells do not get to end by the end of the sample in 2000, making them right censored. For these estimates I only keep complete spells with a clear start and a clear end date.

Figure 4: Quarterly Canadian–non-Canadian differences in the probability of charting and in slot-weeks held



Notes: One observation is an artist $\times$ quarter over 1995Q3–2000Q4. Panels (a) and (c): every artist in the merged MusicBrainz–Discogs catalogue active in at least one year of 1994–1997, active in year t meaning a deduplicated song first released in t , $t - 1$ or $t - 2$ (so at least one first released 1992–1997); 60,663 artists, 1,873 Canadian. Panels (b) and (d): the same definition over 1994–2000 (at least one song first released 1992–2000); 88,955 artists, 2,944 Canadian. In all four, artists carry a country tag and a documented debut year, artists classified as classical or jazz and MusicBrainz special-purpose entities such as **Various Artists** are excluded, and an artist enters at its catalogue debut year. In panels (a)–(b) the outcome is an indicator for the artist appearing on any of the four RPM song charts published without interruption over 1994–2000 (Top 100, Country, Adult Contemporary, Dance) in the quarter, in percentage points; pre-period quarterly rates are 4.20% for Canadian artists and 0.38% for the rest in panel (a) and 4.01% for Canadian artists and 0.36% for the rest in panel (b). In panels (c)–(d) the outcome is the number of chart $\times$ issue slots the artist holds on those charts in the quarter, zero when not charting; pre-period means are 0.437 for Canadian artists and 0.039 for the rest in panel (c) and 0.418 for Canadian artists and 0.037 for the rest in panel (d). Points are the Canadian $\times$ quarter interactions of an OLS regression with artist fixed effects and debut cohort $\times$ quarter fixed effects, a debut cohort being one calendar year of first documented release; observations: 1,274,198 in (a); 1,540,760 in (b); 1,213,541 in (c); 1,451,811 in (d). The hollow point at 1997Q2 is the omitted quarter. Bars are 95% intervals, cluster-robust on the artist; points sit at the quarter's midpoint. Vertical rules mark 1997-08 (review announced), 1998-04 (final report) and 1999-01 (the rule binds). The boxed figures are the same regression with the quarter indicators replaced by three era indicators, in the panel's unit with artist-clustered standard errors, estimated on every quarter. Quarters publishing fewer than 75% of the median issue count (2000Q4, 5 issues against a median of 12) are drawn hollow in panels (a)–(b) and excluded from the quarterly regressions of panels (c)–(d). 1995Q1 and 1995Q2 are excluded: their matched share of published slots is 56% against 77–82% in every other quarter.

Figure 5: Quarterly Canadian–non-Canadian differences in chart-run outcomes



Notes: Each panel plots the Canadian $\times$ quarter coefficients of one regression, the specification of Tables 3 and 4 with the era indicators replaced by quarter indicators: quarter and chart fixed effects, plus artist fixed effects in panels (a)–(g). One observation is an uncensored chart run, dated at the quarter it starts in (2,639–3,160 observations). Window 1995Q3–2000Q4. The hollow point at zero is the omitted quarter, 1997Q2. Bars are 95% intervals, cluster-robust on the artist. Points sit at the quarter’s midpoint. Hollow points mark quarters publishing fewer than 75% of the median issue count (2000Q4, 5 issues, against a median of 12). Excluded from the estimation sample: 2000Q3, 2000Q4 in the run length panel; 2000Q3, 2000Q4 in the start to peak panel; 2000Q3, 2000Q4 in the peak to exit panel. Vertical rules mark 1997-08 (review announced), 1998-04 (final report) and 1999-01 (the rule binds).

and a significant increase in the time to exit the chart. The increase in overall run length is large relative to the pre-expansion mean totalling a 23.3% increase, meanwhile the post-peak stretching of the run implies a 66.8% increase. This means that Canadian artists are charting at comparable positions to non-Canadian artists, but they are staying in the charts for longer, especially in the positions after the peak. This is consistent with broadcasters holding on to the songs that have already proven to be popular, and delaying the entry of new songs.

We can further characterize the charting songs by looking at the differences in the characteristics of both the songs and their artists. Table 4 shows that the average charting song is not more likely to be older when we do not control for artist fixed effects, but using only within-artist variation shows that the average

Table 3: Pooled Difference-in-Differences Estimation of the 1999 CanCon Expansion on Chart-Run Trajectories

CA × era	Entry position		Peak position		Run length		Start→peak		Peak→exit	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Review announced	-0.029 (0.023) [0.204]	-0.046 (0.027) [0.096]	0.010 (0.034) [0.766]	0.022 (0.034) [0.525]	0.574 (1.179) [0.626]	1.454 (1.226) [0.236]	-0.329 (0.498) [0.509]	-0.356 (0.541) [0.511]	0.811 (0.925) [0.381]	1.731 (0.986) [0.080]
Final report	-0.004 (0.026) [0.887]	0.008 (0.029) [0.795]	0.021 (0.035) [0.558]	0.029 (0.037) [0.433]	1.223 (1.502) [0.416]	1.647 (1.797) [0.360]	-0.134 (0.593) [0.821]	-0.106 (0.739) [0.886]	1.309 (1.143) [0.252]	1.726 (1.389) [0.215]
Rule in force	-0.021 (0.023) [0.362]	-0.015 (0.027) [0.581]	0.033 (0.032) [0.303]	0.040 (0.038) [0.292]	2.956 (1.400) [0.035]	2.959 (1.594) [0.064]	-0.729 (0.484) [0.132]	-0.786 (0.610) [0.198]	3.557 (1.133) [0.002]	3.616 (1.358) [0.008]
Pre-period mean, Canadian	0.200		0.688		12.697		6.776		5.414	
Pre-period mean, non-Canadian	0.154		0.697		12.892		6.436		5.620	
Artist fixed effects		X		X		X		X		X
Quarter fixed effects	X	X	X	X	X	X	X	X	X	X
Chart fixed effects	X	X	X	X	X	X	X	X	X	X
Observations	3,168	2,740	3,168	2,740	3,168	2,740	3,168	2,740	3,168	2,740

The sample is every entry on the four RPM song charts published without interruption over 1994–2000 (Top 100, Country, Adult Contemporary, Dance), matched to the merged MusicBrainz–Discogs artist panel, with artists classified as jazz or classical excluded: 230 Canadian and 875 other artists. One observation is an uncensored chart run: a contiguous stretch of issues by one song on any of the four charts, broken by an absence of more than two consecutive issues, dated at the issue in which it begins; runs still on the chart at the last RPM issue (6 November 2000) are dropped. Each column is a separate OLS regression of the outcome on indicators for each era interacted with the credited artist being Canadian, plus the fixed effects marked. The omitted era is 1995Q3–1997Q2, so every coefficient is measured against that period and not against the era before it; review announced is 1997Q3–1998Q1, final report is 1998Q2–1998Q4, and rule in force is 1999Q1–2000Q4. Entry position is the run’s position at its first issue and peak position the best position reached, both expressed as $(\text{depth} - \text{rank} + 1)/\text{depth}$, so 1 is the top slot. Run length is the number of issues in the run, start→peak the issues from its first appearance to its best position, and peak→exit the issues from that position to its last appearance; the two sum to the run’s span. Chart fixed effects are for the chart the song spent most of its weeks on. Standard errors in parentheses are cluster-robust on the artist, the level at which treatment is assigned; bracketed values are p -values from the corresponding t -test. Pre-period means are the within-arm means of the outcome over the omitted era.

Canadian charting song is 0.62 years older than the average non-Canadian one. As well, I find no significant difference in the probability of these songs being the best hit of the artists, or on the career age of the artist when the song enters the charts. I do find a slightly significant decrease in the probability that the charted songs were released during the debut year of the artist, albeit with some nonlinear patterns. This is consistent with broadcasters not only holding on to hits, but also avoiding the rotation of new songs.

We can further decompose the effect on the charting trajectory of the policy. From Figure 2 we know that charting follows a standard hump-shaped trajectory, and it is likely that the policy had an effect within this trajectory. To measure this, I estimated equation 3, looking at the difference between Canadian and non-Canadian artist songs within each week of a charting spell. The results are shown in Figure 6. In line with the aggregate effect on each charting spell, we see a significant increase in the charting position of a song during the first weeks of the spell, which is consistent with the negative sign for the entry to peak position effect. After the first weeks there is no effect whatsoever on the relative position achieved. At the same time, we see no difference between Canadian made songs and non-Canadian ones in the probability of charting during the start of the spell, but we see a significant difference after week 7, lasting up until week 21. This

Table 4: Pooled Difference-in-Differences Estimation of the 1999 CanCon Expansion on the Songs and Artists Charting

CA × era	Song age		Best hit		Debut-year	Career age
	(1)	(2)	(3)	(4)	(5)	(6)
Review announced	-0.210 (0.322) [0.515]	-0.011 (0.269) [0.967]	0.031 (0.030) [0.305]	0.018 (0.038) [0.634]	-0.123 (0.042) [0.003]	-0.926 (0.859) [0.281]
Final report	0.386 (0.298) [0.196]	0.424 (0.373) [0.256]	0.036 (0.029) [0.214]	0.034 (0.032) [0.295]	0.003 (0.057) [0.959]	0.109 (0.859) [0.899]
Rule in force	0.438 (0.274) [0.110]	0.616 (0.249) [0.014]	0.044 (0.029) [0.132]	0.036 (0.034) [0.279]	-0.071 (0.041) [0.088]	-0.313 (0.821) [0.703]
Pre-period mean, Canadian	0.562		0.343		0.156	5.958
Pre-period mean, non-Canadian	0.801		0.379		0.089	7.752
Artist fixed effects	X		X			
Quarter fixed effects	X	X	X	X	X	X
Chart fixed effects	X	X	X	X	X	X
Observations	3,072	2,639	3,168	2,740	3,160	3,160

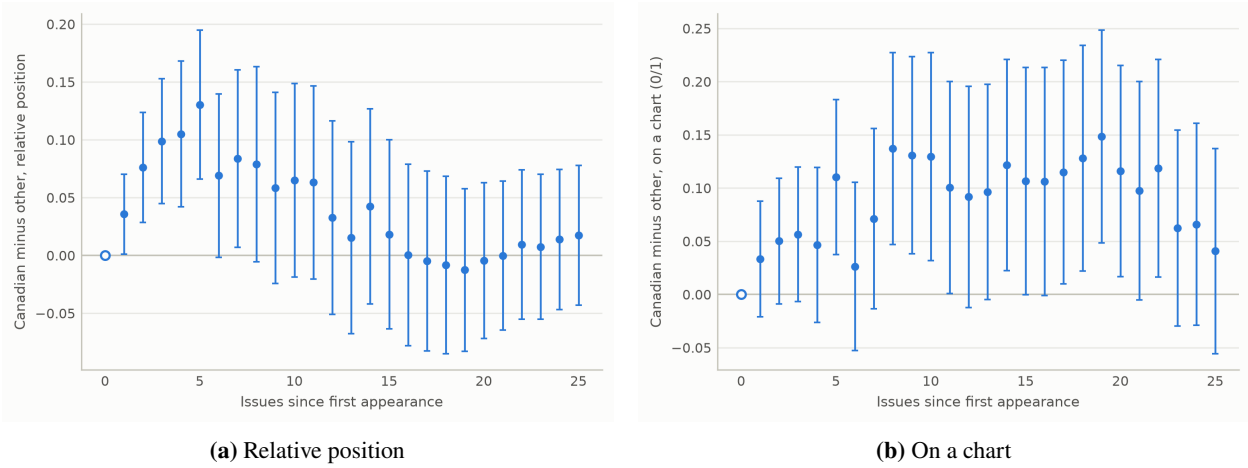
The sample is every entry on the four RPM song charts published without interruption over 1994–2000 (Top 100, Country, Adult Contemporary, Dance), matched to the merged MusicBrainz–Discogs artist panel, with artists classified as jazz or classical excluded: 230 Canadian and 875 other artists. One observation is an uncensored chart run: a contiguous stretch of issues by one song on any of the four charts, broken by an absence of more than two consecutive issues, dated at the issue in which it begins; runs still on the chart at the last RPM issue (6 November 2000) are dropped. Each column is a separate OLS regression of the outcome on indicators for each era interacted with the credited artist being Canadian, plus the fixed effects marked. The omitted era is 1995Q3–1997Q2, so every coefficient is measured against that period and not against the era before it; review announced is 1997Q3–1998Q1, final report is 1998Q2–1998Q4, and rule in force is 1999Q1–2000Q4. Song age is the run’s first issue date minus 1 July of the earliest release year matched to the song, in years, not truncated at zero; it is undefined for the 3.0% of runs below –1 year, treated as release-match failures. Best hit is an indicator for a run of the song with the artist’s highest peak position on the four charts between 10 January 1994 and 30 June 1997, ties broken by slot-weeks over the same span, and is zero for every run of an artist not charting in that span. Debut-year is an indicator for a run whose first issue falls in the calendar year of the artist’s first documented release in the MusicBrainz–Discogs catalogue. Career age is the run’s first issue date minus 1 July of that year, in years. Both are undefined for the 0.3% of runs with no documented debut year or a career age below –1 year. Chart fixed effects are for the chart the song spent most of its weeks on. Standard errors in parentheses are cluster-robust on the artist, the level at which treatment is assigned; bracketed values are p -values from the corresponding t -test. Pre-period means are the within-arm means of the outcome over the omitted era.

explains the increase in the post-peak run length. Noting that the majority of runs last less than 12 weeks, this pattern is increasing the post-peak length of most charting spells.

We can also ask where in the chart these extra weeks are spent. Table 5 splits the weeks of each spell by the third of the chart the song was in at each week, and re-runs equation 2 on each part, so the parts add up to the effects in Table 3. All of the increase in run length comes from the middle and bottom thirds of the chart, with the larger effect in the bottom third, and the same holds for the weeks after the peak. The top third does not move. This adds support to the concentration of the charts, where the lower positions become congested by songs that are kept in the air after their peak, which decreases the room for new songs to enter.

A concern with these results is that maybe these are driven by right-tailed values. For instance, around the

Figure 6: Canadian–non-Canadian difference in the chart trajectory, by issues since a song’s first appearance



Notes: Points are the coefficients θ_k of Equation (3). One observation is a song $\times$ issue, for the first 26 issues after the song’s first appearance on any of the four RPM song charts published without interruption over 1994–2000 (*Top 100*, *Country*, *Adult Contemporary*, *Dance*); a return to the charts within those issues counts where it falls. Panel (a): relative position (depth – rank + 1)/depth at the issue, 0 when off the charts. Panel (b): an indicator for being on any of the four charts. Sample: songs first appearing from 1 July 1995 whose 26 issues end by the last RPM issue (6 November 2000), entering before 1 August 1997 or from 1 January 1999 (last entry 8 May 2000); songs entering in between are excluded. Songs (artists): Canadian 350 (154) before and 160 (90) after; non-Canadian 1,083 (516) before and 531 (319) after; 55,224 observations, 2,124 songs, 886 artists. The chart fixed effect is the chart the song spent most issues on. Issue 0 is omitted (hollow point). Bars are 95% intervals, cluster-robust on the artist. The sum of the coefficients over issues 1–25 is 0.99 (0.75) in panel (a) and 2.31 (0.82) in panel (b), standard errors in parentheses, computed from their joint covariance; the corresponding pre-period sums of the outcome per song are 7.33 for Canadian and 6.86 for non-Canadian songs in panel (a) and 12.80 for Canadian and 12.26 for non-Canadian songs in panel (b).

quota timing both Shania Twain’s *Come On Over* album and Céline Dion released hugely successful albums which could drive some observed differences. To account for this, I reestimate the same equation winsorizing any continuous valued outcome at the 1st and 99th percentiles. The results are shown in Figure B2 and Tables B1 and B2. Overall, the results are of similar magnitudes and signs, meaning these are not driven by extreme realizations and are a more general change.

For a policy that intends to give exposure to more Canadian artists to reach its goal, given that broadcasters rotate songs, we must observe entry of new artists regardless of the broadcaster’s response. If no new artists enter, then the only response is to substitute between existing artists. If the existing artist stock is appealing enough to the broadcaster’s audience, then they would demand music from new artists. To measure this aggregate response of the Canadian record industry to the CanCon expansion, I show the synthetic controls estimates in Figure 7. A general increase in the mandated airplay quota for Canadian artists should be interpreted by potential new entrants as an increase in expected revenues. The previous results show that this might not be the case depending on the overall response of broadcasters to the quota, but it is not clear if individual artists can accurately forecast the broadcaster’s response, plus, it has been documented that artists often choose based more on their own preferences than on market conditions (Caves, 2003). Nonetheless, the estimated synthetic control differences should inform on whether the artist stock in Canada responded to the availability of new mandated airplay slots.

The figure shows a positive deviation that accumulates over time. Using the conformal inference method

Table 5: Decomposition of the Chart-Run Effects by Chart Third

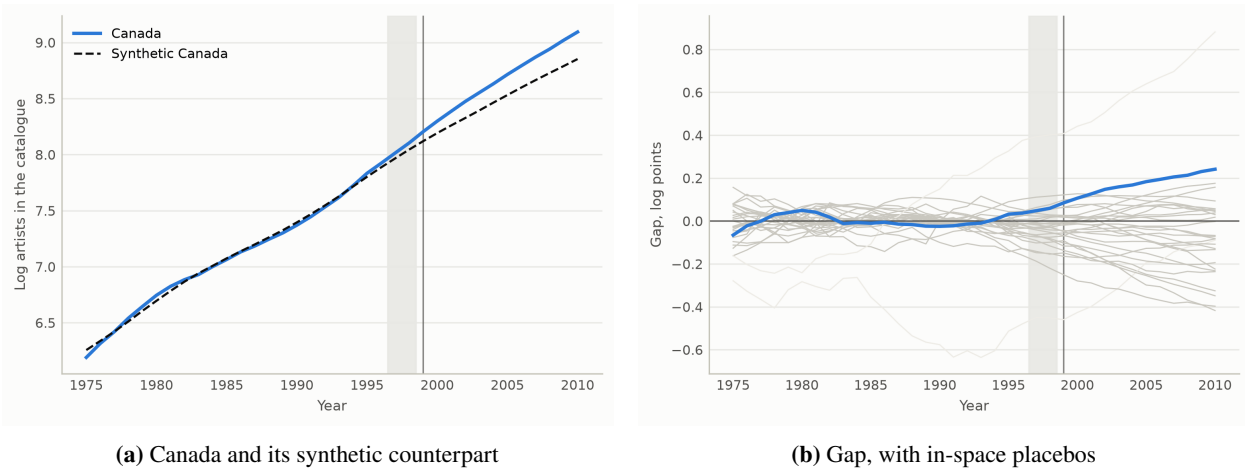
Rule in force $\times$ CA	Run length		Start $\rightarrow$ peak		Peak $\rightarrow$ exit	
	(1)	(2)	(3)	(4)	(5)	(6)
Top third	0.492 (0.842) [0.559]	0.062 (1.092) [0.955]	-0.220 (0.334) [0.510]	-0.472 (0.451) [0.295]	0.650 (0.551) [0.239]	0.469 (0.752) [0.533]
Middle third	1.010 (0.484) [0.037]	1.231 (0.565) [0.030]	-0.154 (0.243) [0.525]	-0.023 (0.297) [0.940]	1.192 (0.376) [0.002]	1.246 (0.439) [0.005]
Bottom third	1.453 (0.509) [0.004]	1.667 (0.591) [0.005]	-0.170 (0.178) [0.340]	-0.157 (0.217) [0.469]	1.658 (0.485) [0.001]	1.897 (0.599) [0.002]
Off the chart			-0.184 (0.048) [0.000]	-0.134 (0.051) [0.008]	0.057 (0.052) [0.279]	0.004 (0.057) [0.941]
Total	2.956 (1.400) [0.035]	2.959 (1.594) [0.064]	-0.729 (0.484) [0.132]	-0.786 (0.610) [0.198]	3.557 (1.133) [0.002]	3.616 (1.358) [0.008]
Artist fixed effects		X		X		X
Quarter fixed effects	X	X	X	X	X	X
Chart fixed effects	X	X	X	X	X	X
Observations	3,168	2,740	3,168	2,740	3,168	2,740

The sample is every entry on the four RPM song charts published without interruption over 1994–2000 (Top 100, Country, Adult Contemporary, Dance), matched to the merged MusicBrainz–Discogs artist panel, with artists classified as jazz or classical excluded: 230 Canadian and 875 other artists. One observation is an uncensored chart run: a contiguous stretch of issues by one song on any of the four charts, broken by an absence of more than two consecutive issues, dated at the issue in which it begins; runs still on the chart at the last RPM issue (6 November 2000) are dropped. Each cell is a separate OLS regression with the specification of Table 3; reported is the rule-in-force (1999Q1–2000Q4) $\times$ Canadian coefficient, against 1995Q3–1997Q2. Columns: run length is the number of issues the run is on the chart; start $\rightarrow$ peak the issues from its first issue to its peak issue, and peak $\rightarrow$ exit from its peak issue to its last. Rows split each outcome into the issues spent in the top, middle and bottom third of the chart, by relative position (depth – rank + 1)/depth in (2/3, 1], (1/3, 2/3] and (0, 1/3], and, for the two spans, the issues inside them on which the song is off the chart. The rows of a column sum to its Total, which is the corresponding coefficient of Table 3. Chart fixed effects are for the chart the song spent most of its weeks on. Standard errors in parentheses are cluster-robust on the artist; bracketed values are p -values from the corresponding t -test.

from Chernozhukov et al. (2021), the difference is not statistically significant. At the same time, when permuting the treatment unit to all the other donor countries, Canada exhibits the largest post-treatment deviation when considering countries with a good pre-treatment fit. In addition, another placebo test performed in a year with no CRTC policy change such as 1981 is shown in Figure B3. The post-treatment deviation is close to 0 on average and the full post-treatment deviation runs through the middle of the donor country distribution. This provides some support for the interpretation that the 1999 CRTC policy change was what drove the divergence measured in Figure 7, but more robust evidence is needed.

A positive deviation relative to the synthetic control is consistent with artists reading an increase in

Figure 7: Synthetic Controls Estimation of the 1999 CanCon Expansion on the Stock of Artists



Notes: The outcome is the log of the number of artists in the merged MusicBrainz–Discogs catalogue whose first release is in or before the year, counted from 1950, by artist country of origin. Artists classified as classical or jazz and artists with no recorded country are excluded. The treated unit is Canada. The donor pool is the 29 countries with at least 1,000 artists in the panel and a debut in every year of the fit window. Weights are non-negative, sum to one, and minimise the squared distance to the Canadian series over 1975–1996; the largest are the United States (0.10), the United Kingdom (0.08) and Germany (0.07), and 26 donors receive more than 0.01. The shaded band marks 1997–1998, excluded from the weight fit, the prediction errors and the test; the vertical line marks 1999. In panel (b) each grey line is the same estimator with one donor as the treated unit, fitted on the other 28; lighter lines are donors whose 1975–1996 root-mean-square prediction error exceeds five times Canada’s, and the United States is not drawn. The mean gap over 1999–2010 is 0.172 log points. The conformal p -value for no effect (Chernozhukov et al., 2021), with the weights re-estimated on the 34-year test block under the null and moving-block permutations of the residuals, is 0.912; the 95% interval for a constant effect is $[-1.006, 2.454]$ log points. The in-space placebo p -value, the share of the 30 units whose ratio of post- to pre-treatment root-mean-square prediction error is at least Canada’s (Abadie, 2021; Abadie et al., 2010), is 0.067. Across 500 resamplings of the donor pool with replacement, the 5th and 95th percentiles of the mean gap are 0.132 and 0.220.

expected revenues from the policy change. However, we cannot rule out that other events moved either the donor stock or Canada. First, the Asian financial crisis of 1997–1998 could still have effects rippling through some of the donor countries. Right after the policy the Dot-com bubble burst, which could have affected the US and UK markets. As well, both the UK and the US are also recipient markets for Canadian artists, so a positive deviation in Canada could also affect them. The current results are implicitly assuming the Canadian contribution is too small to move overall US and UK entry patterns. The fact that over 1994–2000 only 32 Canadian artists are charted in Billboard points to this direction. In any case, the results are not significant, so we cannot reject the null of no increase in artists relative to non-Canadian countries, but the positive sign and null placebo suggest some increase in the Canadian artist stock is possible. An analysis leveraging within country propensity to enter the music market could provide a more complete picture regarding artist entry patterns into recording after a quota increase.

Finally, one of the goals of supporting a nascent industry is to enable the possibility of exporting the output from the protected industry if the market failure that justified the intervention is corrected. If this was the case, we should see a change in the rate at which Canadian artists appear in Billboard charts. Across the Billboard span Canadian artists have frequently achieved high positions, but this only happens for a very select few artists, which are generally the right tail draws of productivity or popularity in their output. A suggestive exercise to measure if this is happening after the 1999 CanCon expansion is shown in Figure A4.

Table 6: RPM Chart-Run Characteristics and Billboard Charting of Canadian Songs, by Era

	Entry position (1)	Peak position (2)	Start→peak (3)	Peak→exit (4)	Sum (5)
<i>Panel A. Charted on Billboard</i>					
Pre-period slope	-0.1311 (0.1318)	0.0646 (0.1518)	-0.0140 (0.0082)	0.0270 (0.0080)	
Post-period slope	-0.0195 (0.1125)	0.1494 (0.1612)	-0.0058 (0.0084)	0.0059 (0.0030)	
Contribution, pre slope	0.0028 (0.0050) [-0.004, 0.016]	0.0021 (0.0076) [-0.011, 0.022]	0.0102 (0.0097) [-0.004, 0.033]	0.0959 (0.0429) [0.026, 0.195]	0.1111 (0.0414) [0.043, 0.206]
Contribution, post slope	0.0004 (0.0037) [-0.009, 0.007]	0.0049 (0.0094) [-0.009, 0.030]	0.0042 (0.0079) [-0.013, 0.021]	0.0211 (0.0143) [-0.001, 0.051]	0.0307 (0.0212) [-0.006, 0.076]
<i>Panel B. Billboard peak position</i>					
Pre-period slope	-0.0907 (0.1071)	0.0350 (0.1183)	-0.0086 (0.0060)	0.0234 (0.0069)	
Post-period slope	0.0000 (0.0851)	0.0698 (0.1069)	-0.0015 (0.0079)	0.0053 (0.0028)	
Contribution, pre slope	0.0019 (0.0038) [-0.004, 0.011]	0.0012 (0.0056) [-0.010, 0.015]	0.0063 (0.0065) [-0.003, 0.022]	0.0833 (0.0377) [0.021, 0.169]	0.0927 (0.0366) [0.032, 0.176]
Contribution, post slope	0.0000 (0.0028) [-0.007, 0.005]	0.0023 (0.0060) [-0.008, 0.017]	0.0011 (0.0072) [-0.015, 0.015]	0.0188 (0.0134) [-0.001, 0.048]	0.0222 (0.0203) [-0.013, 0.064]
Runs, slopes			568		
Bootstrap draws			999		

One observation is an uncensored run on the four RPM song charts (Top 100, Country, Adult Contemporary, Dance) by a Canadian artist, starting in 1995Q3–1997Q2 (pre) or 1999Q1–2000Q4 (post), dated at its first issue. The outcome is the same song on the Billboard R&B/Hip-Hop, Country, Adult Contemporary and Alternative Airplay charts from 26 weeks before the RPM run starts to 10 weeks after it ends: in Panel A an indicator for any appearance, in Panel B its best relative position, 0 if none. Slopes: one OLS regression per panel of the outcome on the four run characteristics of Table 3 and their interactions with a post indicator, with quarter and main-chart fixed effects; standard errors in parentheses cluster-robust on the artist. Contribution: the slope times the channel’s rule-in-force $\times$ Canadian coefficient in Table 3 without artist fixed effects, summed over the four in the last column; standard errors in parentheses and 95% percentile intervals in brackets from a bootstrap that resamples artists with replacement and re-estimates the slopes and the Table 3 coefficients in every draw.

The figure plots a scatter of RPM position weighted chart run lengths against Billboard position weighted chart run lengths. The figure shows a positive correlation between both charting periods, which is expected as both measures capture how popular a song is. However, it is noteworthy how the domain of the post expansion scatter is much more spread, showing Canadian charts giving longer spells to Canadian artists, which sometimes do not even chart in Billboard. A visual calculation suggests that to chart at a similar level in Billboard, a Canadian artist needs 40 position weighted chart weeks after 1999 while it needed around 30 before.

A back of the envelope calculation for this is shown in Table 6. For this I run an OLS regression of US Billboard charting probability and peak position over RPM charting spell characteristics of Canadians for both pre and post expansion charting spells. I then use the estimated coefficients of the expansion of the quota on the Canadian charting spells to predict the expected Billboard charting probability and peak position

before and after the policy. I bootstrap the procedure to produce standard errors for the estimated contribution to Billboard outcomes. The results show that the relationship between Canadian charting length and both Billboard outcomes is weakened after the quota expansion, reflecting that the US market is not responding proportionally to the increased time in the air for Canadian songs. The estimates show that both contributions post expansion are lower than pre-expansion and non-significant. This suggests that the increased airplay found for the quota expansion is not translating into better US charting outcomes.

Summing up, the results point towards a lower probability of securing airplay slots for Canadian artists after the quota expansion, and this is not driven by entry patterns. This is driven by longer charting spells in general and an overall lower number of artists charting. All of the increase in charting length occurs in lower positions of the charts and all of it after a song reaches its peak position. Among the charting artists there is no evidence of higher popularity. Despite this, there is some suggestive evidence that the Canadian artist stock grew after the policy, but it is not possible to disentangle it from other shocks around that time without within country variation. Finally, there is also some evidence that the passthrough from RPM charting spell characteristics to Billboard outcomes has weakened, meaning the reaction to the policy was only driven by broadcaster responses in Canada and either no response from US broadcasters or no major composition change across the set of most appealing songs that could be exported.

6 Conclusion

This paper has tried to understand what were the effects of an expansion of a content quota on artist airplay in Canada. The natural expectation is that a higher quota should increase the exposure consumers get from Canadian artists, and consequently, increase their demand for records. I first find evidence that demand does exhibit both consumption capital dynamics, and satiation, which gives us a distinct lens with which to read any results.

My results point to the strategic response of broadcasters to the policy mainly undermining the objective of increasing Canadian artists' airplay. The quota not only increased the mandated share of airplay but also it increased the stringency of the accepted cases for compliance. This means that broadcasters had to effectively play Canadian music during hours with high demand. A way to avoid losing revenues is to reduce exploration rates and just play the already famous songs again. This is the main result that I find: Canadian artists are more likely to get a longer chart run in Canada, mostly driven by the amount of weeks in the chart after the peak position has been achieved, and by lower ranked songs.

Despite these results, I also found suggestive evidence of an increase in the stock of Canadian artists. This is consistent with artists expecting higher revenues. In future work I will estimate this using within country variation and allowing for entry decisions to jointly determine airplay. Currently, I could only include and exclude debutants, showing that the difference in probability of charting is not driven by debutants, but in the end entry and charting is an endogenous process as only artists that expect to be profitable enter the market.

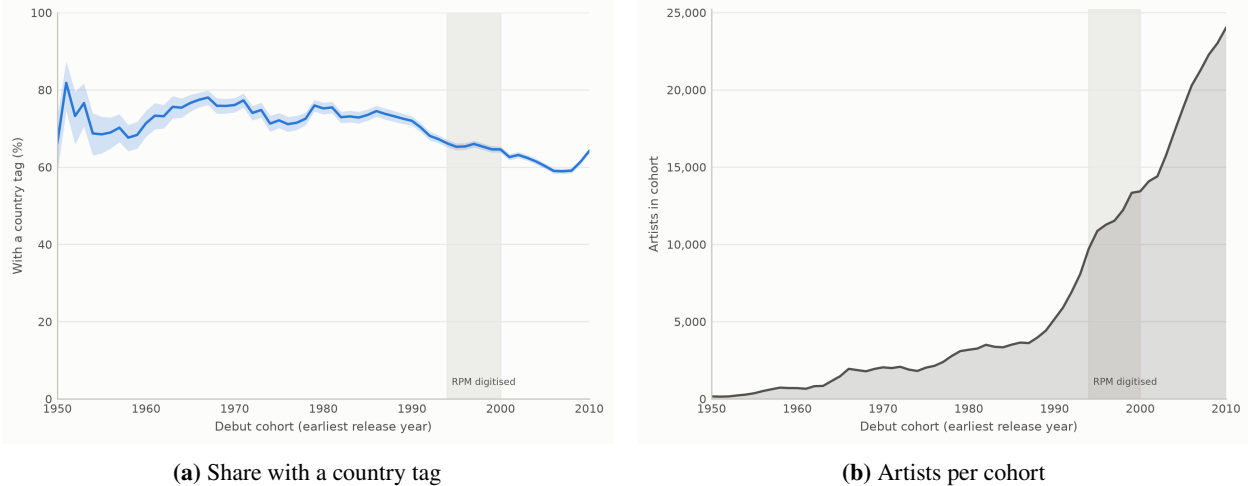
As well, I found suggestive evidence that the longer charting runs in Canada do not drive longer spells in the US. It seems that the selection process for charting in the US is unchanged while charting in Canada became stretched, so artists now chart for longer in Canada and reach similar Billboard positions as before.

All in all, my results highlight the need to consider the incentives and structure of the problem intermediaries face when designing policy. In this case the policy implicitly assumes a neutral allocation from broadcasters, which is not the case, and this directly impacts the effectiveness of it. Potential re-designs that could be considered to induce more entry and an allocation of airplay towards a broader segment of Canadian artists could mimic the French system, which includes a specific quota for new artists. Alternatively, if the objective is entirely about allowing Canadian artists to reach an audience, then funding and promotion mechanisms would be more targeted and would avoid having to forecast the decisions of the broadcasting market altogether.

A main limitation to this project is that all the estimates are over an ordinal aggregate measure of airplay, which are music charts. Procurement of airplay logs would allow me to more precisely characterize the broadcasting response and more broadly characterize how this affects artists. Specifically, it would allow me to determine how this affects smaller and older acts that do not chart but do get airplay, and it would allow me to better quantify the intensive margin of airplay which is currently hidden under the ordinal nature of the data. A main extension to this project is to understand previous expansions of content regulations when the Canadian industry was not as developed as in the 1990s. Therefore, future work is needed to characterize how these effects change if the state of the recording industry is different.

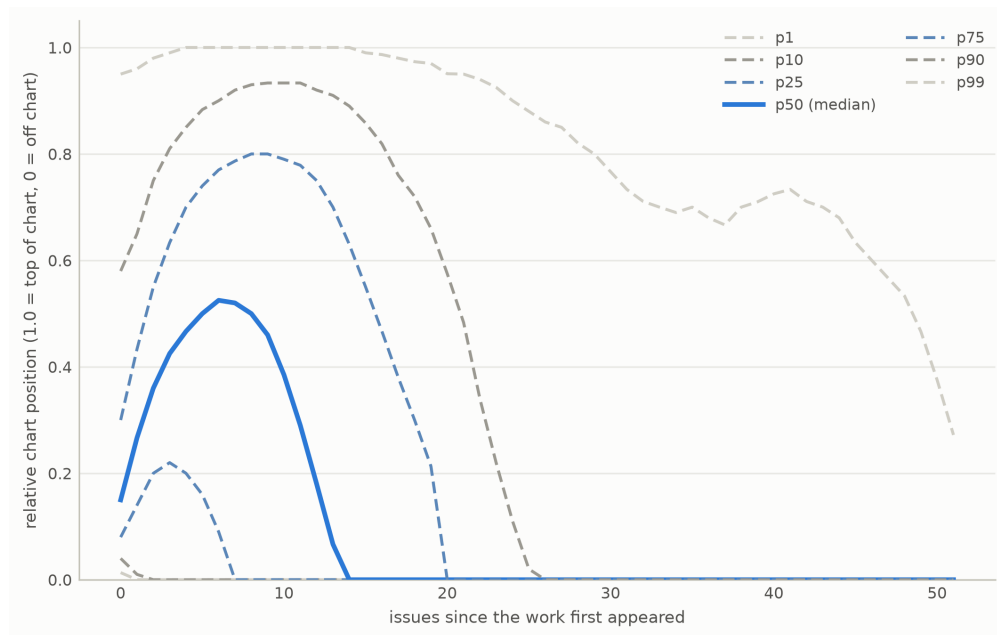
A Additional Descriptives

Figure A1: Artists with country tag by release cohort



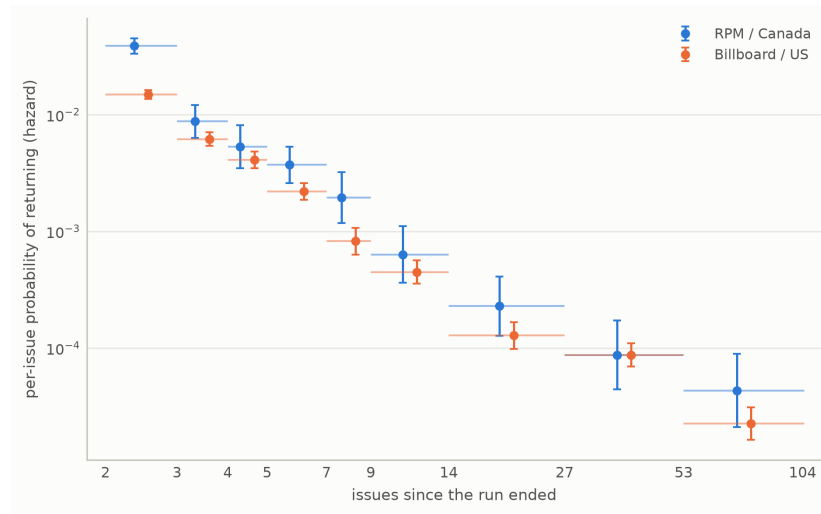
Notes: One observation is a debut cohort: every artist in the merged MusicBrainz–Discogs catalogue is assigned to the earliest release year recorded for them, and panel (a) plots the share of that cohort whose `country` field is non-null. The country tag is the paper’s treatment variable, Canadian against non-Canadian origin; an artist without one is dropped from every specification. Across the whole catalogue 65.1% of 377,323 artists carry a tag. Pooled by decade of debut the share runs 75.9% for the 1960s cohorts, 73.7% for the 1980s and 61.0% for the 2000s, reaching 64.2% in 2010. Over the 1994–2000 cohorts it runs 65.3%. The shaded span in both panels marks those cohorts, the ones the digitised RPM window observes. Panel (b) is the number of artists in each cohort, running from 143 to 24,014 over the plotted range. Bands in panel (a) are 95% Wilson intervals, which assume artists are independent within a cohort. Cohorts before 1950 are omitted from the axis: they hold 2,627 artists, 0.70% of the catalogue, at a tag share of 60.7%. Debut cohort is bounded by catalogue coverage rather than by an artist’s true first release, so an act whose early records never entered MusicBrainz is assigned a later cohort than its real debut.

Figure A2: Chart trajectory of tracks, policy window only



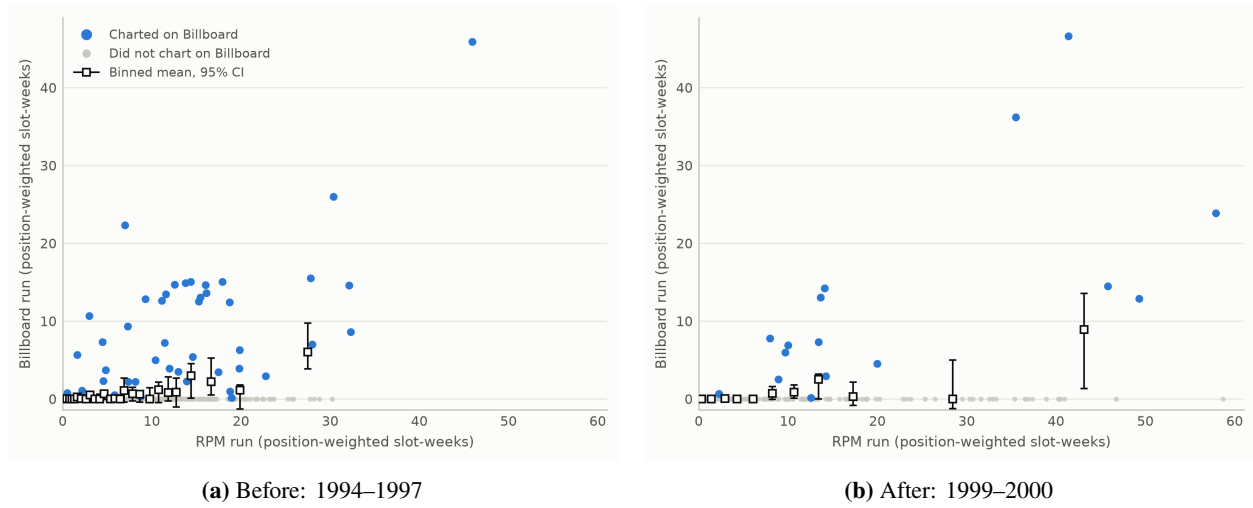
Notes: Percentiles of relative chart position at each issue since a work first appears, pooled across every chart of RPM and Billboard and across the whole 1994–2000 window, with no split by origin, chart or period. The UK Official Charts, which rank sales rather than airplay, and Billboard’s Hot 100, whose methodology changed in December 1998, are excluded. Relative position is $(\text{depth} - \text{rank} + 1) / \text{depth}$, so 1.0 is the top slot of a chart and charts of different depth are commensurable. The cohort is fixed: every work is followed the same 52 issues from its first appearance, and an issue off the chart is encoded as 0, strictly below the lowest available slot, so the set of works behind every percentile is the same at every k . $N = 7,994$ works.

Figure A3: Hazard of returning to the chart after a run ends



Notes: Probability that an ended chart run's work starts a new run in the same market at rest t , given it has not returned before t , over the whole matched span of each market: Billboard / US 1958–2009, 2,674 issues over 4 charts; RPM / Canada 1994–2000, 324 issues over 4 charts. Axes are logarithmic. One observation is a run that ended. A run breaks when the work misses two or more consecutive issues of the market, and a market pools its charts, so the shortest rest is 2 issues and $t = 1$ is not shown. Rest is issues from a run's last appearance to the start of that work's next run in the same market; a work that never returns is censored at the market's last issue. Runs touching either end of the market calendar are dropped. Keeping runs already in progress at the market's first issue, the sample and the share ever returning are RPM / Canada 4,527 runs and 10.8%; Billboard / US 33,839 runs and 4.2%. The hazard is pooled within bins of t (2, 3, 4, 5–6, 7–8, 9–13, 14–26, 27–52, 53–104): returns in the bin over run-issues at risk in the bin. Horizontal segments span each bin; points sit at its geometric centre. Bars are 95% Wilson intervals treating each run-issue at risk as an independent trial. Sample: RPM / Canada 4,150 ended runs by 3,858 works, 7.3% returning within the sample, median rest among those that do 2 issues; Billboard / US 33,813 ended runs by 32,420 works, 4.2% returning within the sample, median rest among those that do 3 issues.

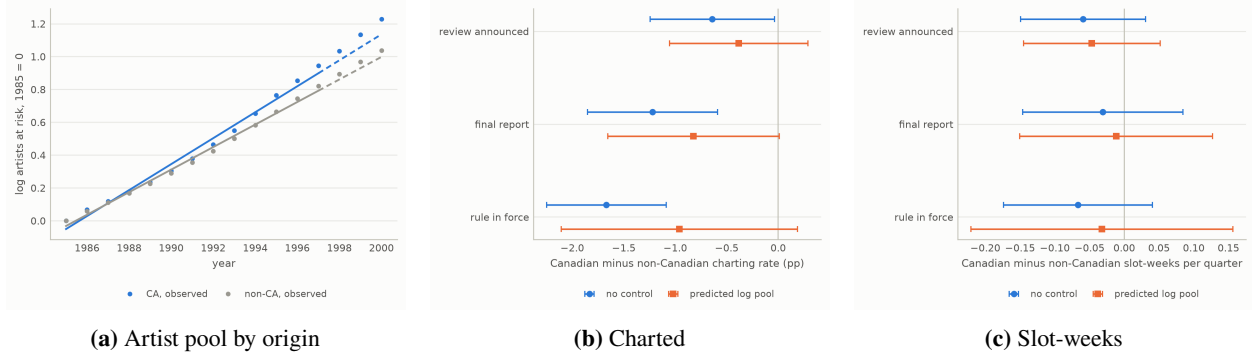
Figure A4: Billboard chart runs against RPM chart runs of Canadian records



Notes: One observation is a song by a Canadian artist, identified by artist and normalised title, with a complete run on the four RPM song charts (Top 100, Country, Adult Contemporary, Dance) inside the window; a run ends after more than two consecutive missed issues, and a song with several runs enters with its first. The horizontal axis is that run's position-weighted slot-weeks: every chart and issue the song occupies counts $(\text{depth} - \text{rank} + 1) / \text{depth}$, 1 for the top slot, summed over the run. The vertical axis is the same quantity for the song's run on the Billboard R&B/Hip-Hop, Country, Adult Contemporary and Alternative Airplay charts with the largest value among those starting between 26 weeks before and 52 weeks after the RPM run starts, and 0 if there is none. Artists classified as jazz or classical are excluded. Windows: 10 January 1994 to 31 July 1997 (panel a) and 1 January 1999 to 6 November 2000 (panel b). Binned means are averages of the vertical-axis value within bins placed at quantiles of the horizontal axis, with the number of bins chosen by the integrated-mean-squared-error-optimal rule of Cattaneo et al. (2024), separately by panel: 26 in panel (a) and 11 in panel (b). Their 95% intervals come from a piecewise-linear fit on the same bins and are cluster-robust on the artist. Panel (a): 544 songs by 181 artists, 43 of them on Billboard. Panel (b): 166 songs by 91 artists, 16 of them on Billboard.

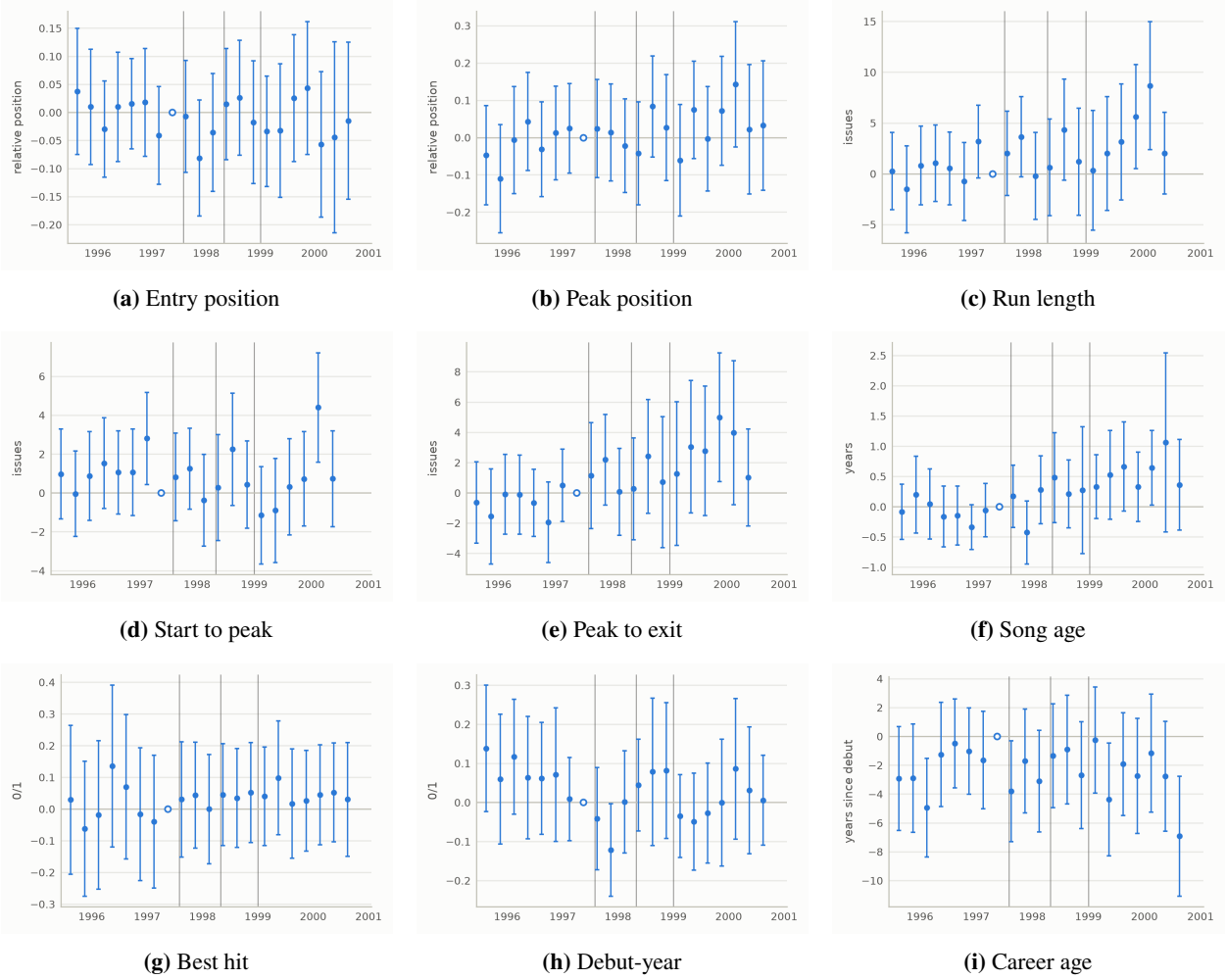
B Robustness Checks

Figure B1: Charting effect controlling for the predicted size of the artist pool



Notes: Panel (a): artists at risk of charting, by origin and year, indexed to 1985. An artist is counted from the year of its first documented release onward; the count covers every artist with a country tag and a documented debut, classical, jazz and MusicBrainz special-purpose entities excluded, and is NOT restricted by the active screen the estimation sample applies — that screen requires activity by 1997 and so holds its own risk set fixed from 1997 on. Markers are observed counts; the solid line is an origin-specific log-linear trend fitted over 1985–1997 and the dashed line is that trend extrapolated. Fitted slopes are 0.0791 for Canadian artists and 0.0687 for the rest, a Canadian-minus-other differential of 0.0104 log points a year. Panels (b) and (c): the Canadian $\times$ era coefficients of Figure 4's regressions with the quarter dummies replaced by the three era dummies, estimated with and without the predicted log pool of panel (a) as a control, with 95% intervals cluster-robust on the artist: panel (b) for charted, in percentage points, as in panel (a) of Figure 4; panel (c) for slot-weeks, in slot-weeks, as in its panel (c). The control is the predicted and not the realised pool, entry after 1997 being itself an outcome; its own coefficient is -20.0 (standard error 15.5) percentage points per log point in panel (b) and -0.97 (standard error 2.46) slot-weeks per log point in panel (c). The control enters the era specification only: it is collinear with a full set of Canadian $\times$ quarter dummies, which span every function of origin and quarter. Because both origins' log pools are fitted log-linearly, the predicted Canadian-minus-other gap is linear in the year, so the control absorbs any Canadian-specific monotone movement over the window, a phased-in effect of the policy included. $N = 1,274,198$ artist-quarters.

Figure B2: Quarterly Canadian–non-Canadian differences in chart-run outcomes, continuous outcomes winsorized



Notes: Each panel plots the Canadian $\times$ quarter coefficients of one regression, the specification of Tables B1 and B2 with the era indicators replaced by quarter indicators: quarter and chart fixed effects, plus artist fixed effects in panels (a)–(g). Every outcome other than best hit and debut-year, which are indicators, is capped at its 1st and 99th percentile, computed once per outcome over both arms. One observation is an uncensored chart run, dated at the quarter it starts in (2,639–3,160 observations). Window 1995Q3–2000Q4. The hollow point at zero is the omitted quarter, 1997Q2. Bars are 95% intervals, cluster-robust on the artist. Points sit at the quarter’s midpoint. Hollow points mark quarters publishing fewer than 75% of the median issue count (2000Q4, 5 issues, against a median of 12). Excluded from the estimation sample: 2000Q3, 2000Q4 in the run length panel; 2000Q3, 2000Q4 in the start to peak panel; 2000Q3, 2000Q4 in the peak to exit panel. Vertical rules mark 1997-08 (review announced), 1998-04 (final report) and 1999-01 (the rule binds).

Table B1: Pooled Difference-in-Differences Estimation of the 1999 CanCon Expansion on Chart-Run Trajectories, Continuous Outcomes Winsorized at the 1st and 99th Percentile

CA × era	Entry position		Peak position		Run length		Start→peak		Peak→exit	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Review announced	-0.029 (0.023) [0.207]	-0.045 (0.027) [0.099]	0.010 (0.034) [0.780]	0.021 (0.034) [0.532]	0.587 (1.175) [0.618]	1.473 (1.221) [0.228]	-0.352 (0.496) [0.478]	-0.411 (0.533) [0.442]	0.823 (0.923) [0.373]	1.746 (0.981) [0.076]
Final report	-0.004 (0.025) [0.887]	0.007 (0.029) [0.794]	0.021 (0.035) [0.558]	0.029 (0.037) [0.430]	1.064 (1.417) [0.453]	1.430 (1.698) [0.400]	-0.181 (0.583) [0.757]	-0.155 (0.730) [0.832]	1.208 (1.052) [0.251]	1.594 (1.291) [0.217]
Rule in force	-0.021 (0.023) [0.364]	-0.015 (0.027) [0.572]	0.033 (0.032) [0.305]	0.039 (0.038) [0.295]	2.500 (1.284) [0.052]	2.547 (1.472) [0.084]	-0.693 (0.479) [0.148]	-0.750 (0.605) [0.215]	3.073 (0.988) [0.002]	3.177 (1.192) [0.008]
Pre-period mean, Canadian	0.200		0.688		12.697		6.770		5.414	
Pre-period mean, non-Canadian	0.154		0.697		12.892		6.421		5.620	
Artist fixed effects		X		X		X		X		X
Quarter fixed effects	X	X	X	X	X	X	X	X	X	X
Chart fixed effects	X	X	X	X	X	X	X	X	X	X
Observations	3,168	2,740	3,168	2,740	3,168	2,740	3,168	2,740	3,168	2,740

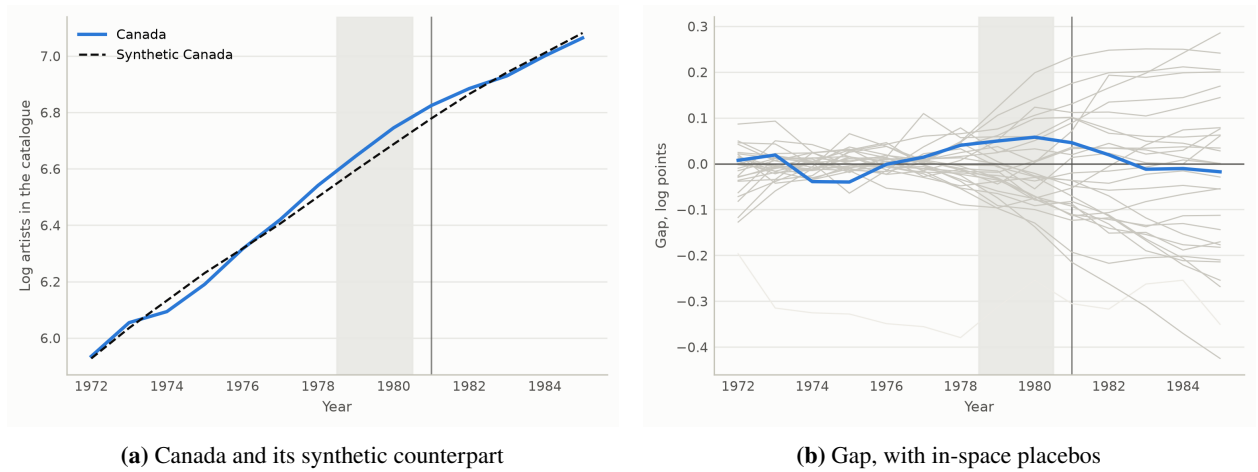
The sample is every entry on the four RPM song charts published without interruption over 1994–2000 (Top 100, Country, Adult Contemporary, Dance), matched to the merged MusicBrainz–Discogs artist panel, with artists classified as jazz or classical excluded: 230 Canadian and 875 other artists. One observation is an uncensored chart run: a contiguous stretch of issues by one song on any of the four charts, broken by an absence of more than two consecutive issues, dated at the issue in which it begins; runs still on the chart at the last RPM issue (6 November 2000) are dropped. Each column is a separate OLS regression of the outcome on indicators for each era interacted with the credited artist being Canadian, plus the fixed effects marked. The omitted era is 1995Q3–1997Q2, so every coefficient is measured against that period and not against the era before it; review announced is 1997Q3–1998Q1, final report is 1998Q2–1998Q4, and rule in force is 1999Q1–2000Q4. Entry position is the run’s position at its first issue and peak position the best position reached, both expressed as (depth – rank + 1)/depth, so 1 is the top slot. Run length is the number of issues in the run, start→peak the issues from its first appearance to its best position, and peak→exit the issues from that position to its last appearance; the two sum to the run’s span. Chart fixed effects are for the chart the song spent most of its weeks on. Standard errors in parentheses are cluster-robust on the artist, the level at which treatment is assigned; bracketed values are *p*-values from the corresponding *t*-test. Pre-period means are the within-arm means of the outcome over the omitted era.

Table B2: Pooled Difference-in-Differences Estimation of the 1999 CanCon Expansion on the Songs and Artists Charting, Continuous Outcomes Winsorized at the 1st and 99th Percentile

CA $\times$ era	Song age		Best hit		Debut-year	Career age
	(1)	(2)	(3)	(4)	(5)	(6)
Review announced	-0.033 (0.201) [0.870]	0.040 (0.196) [0.839]	0.031 (0.030) [0.305]	0.018 (0.038) [0.634]	-0.123 (0.042) [0.003]	-0.921 (0.853) [0.280]
Final report	0.377 (0.217) [0.082]	0.408 (0.268) [0.128]	0.036 (0.029) [0.214]	0.034 (0.032) [0.295]	0.003 (0.057) [0.959]	0.152 (0.849) [0.858]
Rule in force	0.373 (0.165) [0.024]	0.618 (0.220) [0.005]	0.044 (0.029) [0.132]	0.036 (0.034) [0.279]	-0.071 (0.041) [0.088]	-0.247 (0.811) [0.761]
Pre-period mean, Canadian	0.511		0.343		0.156	5.962
Pre-period mean, non-Canadian	0.725		0.379		0.089	7.730
Artist fixed effects		X		X		
Quarter fixed effects	X	X	X	X	X	X
Chart fixed effects	X	X	X	X	X	X
Observations	3,072	2,639	3,168	2,740	3,160	3,160

The sample is every entry on the four RPM song charts published without interruption over 1994–2000 (Top 100, Country, Adult Contemporary, Dance), matched to the merged MusicBrainz–Discogs artist panel, with artists classified as jazz or classical excluded: 230 Canadian and 875 other artists. One observation is an uncensored chart run: a contiguous stretch of issues by one song on any of the four charts, broken by an absence of more than two consecutive issues, dated at the issue in which it begins; runs still on the chart at the last RPM issue (6 November 2000) are dropped. Each column is a separate OLS regression of the outcome on indicators for each era interacted with the credited artist being Canadian, plus the fixed effects marked. The omitted era is 1995Q3–1997Q2, so every coefficient is measured against that period and not against the era before it; review announced is 1997Q3–1998Q1, final report is 1998Q2–1998Q4, and rule in force is 1999Q1–2000Q4. Song age is the run’s first issue date minus 1 July of the earliest release year matched to the song, in years, not truncated at zero; it is undefined for the 3.0% of runs below –1 year, treated as release-match failures. Best hit is an indicator for a run of the song with the artist’s highest peak position on the four charts between 10 January 1994 and 30 June 1997, ties broken by slot-weeks over the same span, and is zero for every run of an artist not charting in that span. Debut-year is an indicator for a run whose first issue falls in the calendar year of the artist’s first documented release in the MusicBrainz–Discogs catalogue. Career age is the run’s first issue date minus 1 July of that year, in years. Both are undefined for the 0.3% of runs with no documented debut year or a career age below –1 year. Chart fixed effects are for the chart the song spent most of its weeks on. Standard errors in parentheses are cluster-robust on the artist, the level at which treatment is assigned; bracketed values are p -values from the corresponding t -test. Pre-period means are the within-arm means of the outcome over the omitted era.

Figure B3: Synthetic Controls Estimation with a Placebo Policy Year of 1981



Notes: The outcome, estimator, donor pool, inference and drawing conventions are those of Figure 7, with the treatment date moved to 1981. Weights are fitted on 1972–1978, the shaded band marks 1979–1980, excluded from the weight fit, the prediction errors and the test, and the post-treatment window is 1981–1985. The mean gap over 1981–1985 is 0.005 log points. The conformal p -value for no effect, on the 12-year test block, is 0.667, and the 95% interval for a constant effect is $[-4.218, 4.246]$ log points. The in-space placebo p -value is 0.867.

C Institutional Setting

This appendix documents the regulatory history of Canadian radio: the statutory basis of the regulator, the Canadian content regime, the 1997–98 review, the rules governing competition between stations, and support for recorded music. Table C1 collects the chronology.

C.1 Regulator and statutory basis

Federal jurisdiction over broadcasting dates from the *Radiotelegraph Act* of 1913 (CCF, 2024). The *Canadian Radio Broadcasting Act* of 1932 created the first regulator, the Canadian Radio Broadcasting Commission, and its 1933 amendments capped foreign programming at 40 per cent (CCF, 2024; Canadian Radio Broadcasting Act, 1932). The *Broadcasting Act* of 1936 transferred regulation of private stations to the newly created Canadian Broadcasting Corporation (Broadcasting Act, 1936). The *Broadcasting Act* of 1958 created the Board of Broadcast Governors (BBG) and capped foreign ownership at 25 per cent (Broadcasting Act, 1958; CCF, 2024). In 1960 the BBG set the first Canadian content quotas, for television only: 45 per cent by April 1961 and 55 per cent by April 1962 (CCF, 2024). The *Broadcasting Act* of 1968 replaced the BBG with the Canadian Radio-Television Commission and brought cable under federal regulation; the Commission took on telecommunications in 1976 and became the Canadian Radio-television and Telecommunications Commission (CRTC) (Broadcasting Act, 1968; Canadian Radio-television and Telecommunications Commission Act, 1976).

The statute governing the period studied is the *Broadcasting Act* of 1991 (Broadcasting Act, 1991). Section 3(1)(a) requires the system to be “effectively owned and controlled by Canadians”; section 3(1)(f) requires each undertaking to “make maximum use . . . of Canadian creative and other human resources”, the basis of the content quota. Sections 9 and 10 give the Commission two instruments: conditions of licence attached to an individual station, and regulations applying to all stations of a class.

C.2 Canadian content regime for radio

C.2.1 MAPL system

Canadian content in music is defined by the MAPL system, adopted with the 1971 quota. A selection qualifies if it meets at least two of four conditions (CRTC, 2022b):

M — Music

the music is composed entirely by a Canadian;

A — Artist

the music or lyrics are performed principally by a Canadian;

P — Performance

the selection is recorded wholly in Canada, or performed wholly in Canada and broadcast live in Canada;

L — Lyrics

the lyrics are written entirely by a Canadian.

Recordings made before January 1972 qualify on any one condition, and instrumental performances of Canadian-composed works qualify on M alone. For recordings made after 1 September 1991, a work co-written with a non-Canadian satisfies M and L if the Canadian receives at least half of both the music and lyric credits, as registered with a performing rights society. “Canadian” means a citizen, a permanent resident, or a person resident in Canada for the preceding six months (CRTC, 2022b). Eligibility is a property of the work at the time of recording, determined by rule from its credits.

C.2.2 Quota levels and scheduling

The quota was introduced for AM radio by SOR/70-256, announced on 22 May 1970 and in force on 18 January 1971, at 30 per cent of musical selections (CCF, 2024; Radio (A.M.) Broadcasting Regulations, 1970). Spalding (2016) documents the proceeding. From 1975 each FM licensee filed a binding Promise of Performance declaring its format and its Canadian content level, which ranged between roughly 10 and 30 per cent (CRTC, 1975). The *Radio Regulations, 1986* (SOR/86-982) consolidated radio regulation; section 2.2 carries the content rules (Radio Regulations, 1986).

Public Notice CRTC 1990-111 raised the FM popular-music (category 2) minimum to 30 per cent, restructured FM format groups into Pop/Rock/Dance, Country and Specialty, and revised the hits policy, which limited English-language FM stations to under 50 per cent charted hits and did not count a Canadian selection as a hit for twelve months after it charted (CRTC, 1990). Public Notice CRTC 1991-89 implemented these changes from 1 September 1991 and removed a separate 5 per cent Canadian-composed requirement (CRTC, 1991). Public Notice CRTC 1995-60 released commercial FM stations from their format commitments from 1 September 1995, and retained the content requirements, the hits policy, and the one-AM-plus-one-FM ownership limit (CRTC, 1995).

Immediately before 1999 the rules were: 30 per cent of category 2 selections Canadian over the broadcast week, for AM and FM commercial stations; 65 per cent French-language vocal music for French-language stations; and an expectation, enforced through licence monitoring rather than regulation, of 25 per cent Canadian content between 6:00 a.m. and 7:00 p.m. on weekdays (CRTC, 1991, 1997a, 1998b).

C.3 1997–98 policy review

Public Notice CRTC 1997-104 of 1 August 1997 opened the review, asking whether to raise the Canadian music minimum, modify MAPL, and relax the common ownership limit. Comments were due 30 September and 3 November 1997, and the hearing was held on 1 December 1997 in Hull, Quebec (CRTC, 1997a). The Commission published its determination, *Commercial Radio Policy 1998*, as Public Notice CRTC 1998-41 of 30 April 1998 (CRTC, 1998b). Proposed regulations followed on 30 July 1998 (PN 1998-80), and the final amendments, SOR/98-597, were published on 17 December 1998 (PN 1998-132) and came into force on 3 January 1999 (CRTC, 1998c,d; Regulations Amending the Radio Regulations, 1986, 1998). The paper’s three event dates follow this sequence: August 1997 (review announced), April 1998 (determination published), and January 1999 (rule in force).

The amendments changed four requirements (PN 1998-41; SOR/98-597):

1. Weekly level: the category 2 minimum for commercial AM and FM stations rose from 30 to 35 per cent (paras. 11, 95).
2. Scheduling: at least 35 per cent of category 2 selections between 6:00 a.m. and 6:00 p.m., Monday to Friday, must be Canadian (paras. 13, 106), replacing the 25 per cent expectation over 6:00 a.m.–7:00 p.m.
3. Counting: only selections played in their entirety count toward either minimum (paras. 13, 108). Oldies stations (90 per cent pre-1981 music) could apply to remain at 30 per cent (CRTC, 1998d).
4. French-language vocal music: the 65 per cent weekly minimum was kept, and a 55 per cent minimum added over the same weekday daypart (paras. 16, 169).

The weekday daytime floor therefore rose from 25 to 35 per cent over a window one hour shorter, and became equal to the weekly floor. In the notice the Commission stated that the previous rule “can lead to lower levels of Canadian content during those periods when most people are listening” (para. 105); its music use study found 28 per cent Canadian content overnight on five English-language Toronto stations and 38 per cent on five French-language Montréal stations (para. 121); and broadcasters argued that in morning drive “addition or deletion of even one record could make a major change to Canadian content levels” (para. 101) (CRTC, 1998b).

The Commission left MAPL unchanged (paras. 15, 146). It did not raise the quota further, recording instead an expectation that voluntary initiatives would bring Canadian content to 40 per cent within five years (para. 12) (CRTC, 1998b). The requirement remains 35 per cent (CRTC, 2006, 2022a, 2024b).

C.4 Competitive environment

C.4.1 Frequency allocation

A station requires a broadcasting licence from the CRTC and a broadcasting certificate for its frequency from the Minister of Industry (now ISED) under the *Radiocommunication Act* (Broadcasting Act, 1991; ISED, 2024c; Radiocommunication Act, 1985). Broadcast spectrum is not auctioned (ISED, 2011). Where several applicants seek one frequency, the CRTC holds a comparative hearing, announced in the *Canada Gazette* at least 60 days ahead, and selects on ownership, financial and technical capacity, and programming, with evidence on market demand, diversity, and the effect on incumbents. The process takes 8 to 18 months, no payment is made for the frequency, and the Commission notes that frequencies are scarce in some major urban areas (CRTC, 2024c).

C.4.2 Entry

Until 1998 the Radio Market Policy screened applications for new stations against group profitability, station profitability, and market revenue growth (para. 75). PN 1998-41 revoked it in favour of assessment on the merits of each application, and opened AM-to-FM conversions to case-by-case review (para. 78) (CRTC, 1998b). By 2005 about 55 per cent of new FM stations were AM conversions (CRTC, 2006).

C.4.3 Common ownership

Before 1998 an owner was generally limited to one AM and one FM station per language per market (CRTC, 1995, 1997a). PN 1998-41 replaced this with two tiers (para. 38) (CRTC, 1998b):

- markets with fewer than eight commercial stations: up to three stations in a language, at most two in one band;
- markets with eight or more commercial stations: up to two AM and two FM stations in a language.

The Commission cited radio's difficulty competing with other media for advertising revenue (para. 26). Applicants must address the effect on news diversity and competition, and equity in other stations and local media in the same market counts toward the limit (paras. 40–41). The Commission rejected the Canadian Association of Broadcasters' more permissive proposal, noting that under it only Toronto, Vancouver, Edmonton and Winnipeg would have more than two owners in any one language (para. 36) (CRTC, 1998b). The limits were next changed in December 2022, to four stations, at most three in one band, in markets of eight or more (CRTC, 2022a).

The *Radio Regulations, 1986* define a station's market as the smaller of its coverage contour (the AM daytime 15 mV/m contour, or the FM 3 mV/m contour) and the central area defined by the Bureau of Broadcast Measurement (CRTC, 2010; *Radio Regulations, 1986*). The contour is computed from transmitter power, antenna height and pattern, frequency, and, for AM, ground conductivity, using ISED's propagation curves (ISED, 2024a,b). The central area is a ratings geography drawn by BBM (now Numeris), typically a Census Metropolitan Area or Census Agglomeration (Numeris, 2024).

C.4.4 Foreign ownership

Foreign control has been restricted since 1958 (Broadcasting Act, 1958). In the period studied, SOR/97-192 of 8 April 1997 bars the issue, amendment or renewal of a licence for a non-Canadian applicant, including one controlled by a non-Canadian through personal, financial, contractual or business relations (Direction to the CRTC (Ineligibility of Non-Canadians), 1997).

C.4.5 Local management agreements

Under a local management agreement (LMA) one licensee manages another station in the same or an adjacent market. PN 1998-41 required multi-station applicants to undertake not to enter new LMAs without prior approval (paras. 45–46), and PN 1998-42 called for comments on LMA policy (CRTC, 1998a,b). PN 1999-55 proposed a review mechanism, enacted by SOR/2000-239 (CRTC, 1999; *Radio Regulations, 1986*).

C.4.6 Transfers and tangible benefits

The Commission approves changes of control without a competitive hearing, subject to a tangible benefits contribution. PN 1998-41 lowered the expected contribution from about 10 per cent to a minimum of 6 per cent of the transaction value, waived for undertakings unprofitable on a three-year average (paras. 69–70).

The 6 per cent is allocated as 3 per cent to a new Canadian music marketing and promotion fund, 2 per cent to FACTOR or MUSICACTION, and 1 per cent to eligible initiatives at the purchaser’s discretion (para. 72) (CRTC, 1998b).

C.4.7 Format regulation

The Promise of Performance bound each FM station to a declared format from 1975 (CRTC, 1975); from 1 September 1995 the Commission generally released stations from those commitments (CRTC, 1995). The hits policy (above) was revised in PN 1997-42 (CRTC, 1997b); French-language stations were exempt from it (CRTC, 1990).

C.5 Support for Canadian recorded music

FACTOR (Foundation Assisting Canadian Talent on Recordings) was founded in 1982 by CHUM, Rogers and Moffat together with the Canadian Independent Record Production Association and the Canadian Music Publishers Association. In 1986 it began administering the federal Sound Recording Development Program, now the Canada Music Fund (CRTC, 2024a; Spalding, 2008). MUSICACTION is its French-language counterpart. The 3 per cent promotion share of transfer benefits funded the Radio Starmaker Fund, established in autumn 2000, with Fonds RadioStar for French-language artists (CRTC, 1998b, 2024a; Radio Starmaker Fund, 2024).

C.6 After 1999

Commercial Radio Policy 2006 (PN 2006-158) reviewed the 1998 policy (CRTC, 2006). It reports that commercial radio revenues grew 5.5 per cent a year over 2001–2005, profit before interest and taxes rose from \$171 million (16 per cent margin) to \$277 million (20.8 per cent), and weekly listening fell by about 1 hour 25 minutes between 1999 and 2005, to 19.1 hours per capita. The Commission attributed the profitability gains to ownership consolidation, and stated that its studies had “not indicated that there has been a meaningful increase in the airplay of selections by new and emerging Canadian artists by English-language stations.” It retained 35 per cent, raised category 3 minimums to 25 per cent for concert music and 20 per cent for jazz and blues, renamed Canadian talent development as Canadian content development, scaled contributions to revenue, and directed at least 60 per cent of basic contributions to FACTOR or MUSICACTION. The *Revised Commercial Radio Policy* (CRTC 2022-332) retained the 35 per cent weekly and daytime minimums (CRTC, 2022a). The *Online Streaming Act* (S.C. 2023, c. 8) extends the *Broadcasting Act* to online undertakings (Online Streaming Act, 2023).

Table C1: Regulatory chronology: Canadian broadcasting and music content, 1913–2023

Date	Instrument	Content
<i>Panel A. Statutory and institutional foundations</i>		
1913	<i>Radiotelegraph Act</i>	First federal jurisdiction over broadcasting.

continued overleaf

Table C1, continued

Date	Instrument	Content
1932	<i>Cdn. Radio Broad-casting Act</i>	Creates the CRBC; 1933 amendments cap foreign programming at 40%.
1936	<i>Broadcasting Act</i>	Dissolves CRBC; CBC regulates private stations.
1958	<i>Broadcasting Act</i>	Creates the Board of Broadcast Governors; 25% foreign ownership cap.
1960–62	BBG rulings	Television Canadian content: 45% by 1 Apr 1961, 55% by 1 Apr 1962. Radio not covered.
1968	<i>Broadcasting Act</i>	Creates the CRTC (in force 1 Apr 1968); cable brought under federal regulation.
1976	<i>CRTC Act</i>	Telecommunications jurisdiction added (1 Apr 1976).
1991	<i>Broadcasting Act</i>	S.C. 1991, c. 11. Current statute; ss. 3(1)(a), 3(1)(f), 9, 10.
2023	<i>Online Streaming Act</i>	S.C. 2023, c. 8. Extends the Act to online undertakings.
<i>Panel B. Content quotas for radio music</i>		
22 May 1970	SOR/70-256	30% Canadian content for commercial AM announced; in force 18 Jan 1971. MAPL adopted.
20 Jan 1975	FM policy	Promise of Performance; FM content set station by station (c. 10–30%).
1986	SOR/86-982	<i>Radio Regulations, 1986</i> ; s. 2.2 carries the content rules.
17 Dec 1990	PN 1990-111	FM raised to 30%; format groups restructured; hits policy revised (Canadian selections not counted as hits for 12 months).
30 Aug 1991	PN 1991-89	Implements the FM policy (in force 1 Sep 1991); 5% Canadian-composed requirement removed.
1 Sep 1991	MAPL amendment	Co-writes qualify on M and L where the Canadian receives $\geq$ half of both credits.
21 Apr 1995	PN 1995-60	Format commitments relaxed from 1 Sep 1995; 30% / 65% and one-AM-one-FM retained.
<i>Panel C. 1997–98 review</i>		
1 Aug 1997	PN 1997-104	Review announced; comments due 30 Sep and 3 Nov 1997.
1 Dec 1997	Public hearing	Hull, Quebec.
30 Apr 1998	PN 1998-41	<i>Commercial Radio Policy 1998</i> . Quota to 35% weekly; new 35% for 6 a.m.–6 p.m. Mon–Fri; entire-play counting rule; French vocal 55% over the same daypart; common ownership relaxed; Radio Market Policy revoked; benefits cut to 6%. MAPL unchanged.
30 Jul 1998	PN 1998-80	Proposed regulations.
17 Dec 1998	PN 1998-132 / SOR/98-597	Final amendments. In force 3 Jan 1999.
<i>Panel D. After 1999</i>		
Autumn 2000	Radio Starmaker Fund	Funded by the 3% marketing-and-promotion share of transfer benefits; Fonds RadioStar for French.
2000	SOR/2000-239	Local management agreements brought under regulation (following PN 1998-42, PN 1999-55).
15 Dec 2006	PN 2006-158	35% retained; CTD becomes CCD; category 3 minimums raised.
7 Dec 2022	CRTC 2022-332	35% weekly and peak retained; ownership cap raised to four stations in markets of 8+.

Notes: “PN” denotes a CRTC Public Notice; “SOR” a Statutory Order and Regulation. Percentages for radio refer to content category 2 (popular music) unless stated. Sources, by panel: Panel A, (Broadcasting Act, 1936, 1958, 1968, 1991; CCF, 2024; Canadian Radio Broadcasting Act, 1932; Canadian Radio-television and Telecommunications Commission Act, 1976; Online Streaming Act, 2023); Panel B, (CRTC, 1990, 1991, 1995, 2022b, 1975; Radio (A.M.) Broadcasting Regulations, 1970; Radio Regulations, 1986); Panel C, (CRTC, 1997a, 1998b,c,d; Regulations Amending the Radio Regulations, 1986, 1998); Panel D, (CRTC, 1999, 2006, 2022a, 2024a; Radio Starmaker Fund, 2024).

Table D1: Data sources

Source	Unit	Span used	How the data are generated
MusicBrainz	Release, track, recording, work, artist	Full catalogue (releases 1900–2010)	Volunteer editors submit and correct entries; edits are peer-voted; core data are public domain
Discogs	Physical/digital release, credit, genre	Full catalogue	Volunteer contributors, many of them sellers on the site’s marketplace; entries peer-voted; monthly public dumps
RPM	Weekly chart entry (rank $\times$ chart $\times$ issue)	1994–2000	Canadian trade weekly; singles-side charts tabulate reported station airplay, album chart tabulates retail
Billboard, Official Charts	Weekly chart entry	Full history (1956/1952 onward)	Monitored airplay and point-of-sale scanning (US, from 1991); electronic point-of-sale panel (UK, from 1994)
RIAA, Music Canada	Certification award (record $\times$ tier)	Full history (1958/1975 onward)	Voluntary, rights-holder-initiated, third-party-audited awards against net shipments (physical era)

Notes: “Span used” is the range retained in the constructed data. Chart and certification archives were retrieved from each publisher’s public archive; MusicBrainz and Discogs are distributed as bulk database dumps.

D Data Appendix

This appendix documents the provenance of each source (Section D.1) and the construction of the catalogue, the treatment variable, and the outcome panels (Section D.2). Table D1 summarises the sources.

D.1 Sources

D.1.1 MusicBrainz

MusicBrainz is an open music encyclopaedia maintained by the non-profit MetaBrainz Foundation (MusicBrainz, 2026a). Registered volunteers create and correct entries by hand; each change is an edit that other editors can inspect and vote on. Edits by established editors and low-risk edit types are applied automatically, the rest after a voting period. Voting requires an account older than two weeks, a verified address, and accepted edits (MusicBrainz, 2026b,c). Entry follows written style guidelines (MusicBrainz, 2026f). Core data are released under CC0 (MusicBrainz, 2026d). Entries are created retrospectively, so the dump records the catalogue as entered up to its date.

The schema separates levels of musical identity (MusicBrainz, 2026e). A work is an abstract composition and carries composer and lyricist relations. A recording is a performance captured in audio. A release is an issued product (a pressing, edition or reissue) within a release group (the album), and a track is the appearance of a recording on one medium of one release. Artists link to each level through typed relationships (composer, lyricist, member of band, recorded at), which supply the credits used to score MAPL.

D.1.2 Discogs

Discogs was founded in 2000 as a discography of electronic music, opened to all genres in 2003, and added a marketplace for physical records in 2005 (Discogs, 2026a). An item can be listed for sale only against an existing database entry, and entries correspond to specific pressings, with label, catalogue number, country, matrix codes, and sleeve credits. Submissions are peer-rated, and contributors with poor ratings face submission limits (Discogs, 2026b,c). Genre is a fixed vocabulary of fifteen top-level genres, each subdivided into styles (Discogs, 2026d). Monthly XML dumps are released under CC0 (Discogs, 2026e).

In this project Discogs supplies composer, lyricist and writer credits, release country, and genre and style. Credits are attached to a release rather than a track, and role names are free text.

D.1.3 RPM

RPM was the Canadian record industry's trade weekly, founded by Walt Grealis and Stan Klees and published from 24 February 1964 to 13 November 2000. It published the national charts for most of that span. The run is digitised as page images and publicly archived (LAC, 2024; World Radio History, 2024).

RPM compiled its charts from record company reports, radio station playlists, and record store reports. From September 1988 the flagship singles chart was tabulated from reporting stations' playlists only, and was later renamed to rank tracks rather than commercial singles (Canadian Encyclopedia, 2024b; RPM's Pop Charted Songs, 2018). Over 1994–2000 the singles-side charts (*RPM 100 Hit Tracks*, *Adult Contemporary*, *Country Tracks*, *Dance*, *Alternative*, *Rock*, *Urban*) rank records by reported airplay across the reporting stations. The issues do not list the reporting stations.

The album chart ranks retail. Nielsen SoundScan began tracking Canadian point-of-sale in November 1996. The 1997 issues credit SoundScan on the album page, print the chart as *Top Selling Albums* (against *RPM 100 Albums* before and *RPM Top 100 CDs* after), and use a different layout; other years describe a ranking from retail reports and airplay. I treat 1997 as a separate regime on the album chart.

D.1.4 Foreign charts: Billboard and the Official Charts Company

Billboard (United States) Before 1991 the *Hot 100* and album chart combined ranked reports from retailers and radio stations. The album chart adopted Nielsen SoundScan point-of-sale data in May 1991, and on 30 November 1991 the *Hot 100* adopted SoundScan together with Broadcast Data Systems monitored airplay (Billboard, 2016). Since 1992, titles below a rank threshold after a set number of weeks (below No. 50 after 20 weeks) move to a recurrent chart (Billboard, 2026). From 5 December 1998 airplay-only titles are eligible for the *Hot 100*, which previously required a commercially available single (Billboard, 2018). Billboard's *Canadian Hot 100* and *Canadian Albums* begin in June 2007 (Canadian Encyclopedia, 2024b).

Official Charts Company (United Kingdom) The UK chart ranks sales only (OCC, 2026). The British Market Research Bureau compiled it from 1969 from a panel of about 250 shops; Gallup took over in 1983 and extended it to 100 positions; the Chart Information Network took responsibility in 1990 (renamed the Official UK Charts Company in 2001). From 1 February 1994 Millward Brown compiled it from electronic

Table D2: Certification thresholds, United States and Canada

	Period	Gold	Platinum	Diamond
<i>Panel A. United States (RIAA)</i>				
Albums	from 1976	500,000	1,000,000	10,000,000
Singles	to late 1980s	1,000,000	2,000,000	–
Singles	from late 1980s	500,000	1,000,000	10,000,000
<i>Panel B. Canada (CRIA / Music Canada)</i>				
Albums	1975–Apr 2008	50,000	100,000	1,000,000
Albums	from May 2008	40,000	80,000	800,000
Singles	to Sep 2002	50,000	100,000	1,000,000
Singles	from Sep 2002	5,000	10,000	100,000
Digital downloads	Mar 2006–Apr 2008	20,000	40,000	400,000
Digital downloads	from May 2008	40,000	80,000	800,000
Single Award	from Sep 2016	40,000	80,000	800,000

Notes: Units per award tier. US Gold was introduced in 1958 (originally a dollar-value criterion), Platinum in 1976, Multi-Platinum in 1984 and Diamond in 1999; the Canadian programme began in 1975. Multi-Platinum tiers are integer multiples of Platinum. The Canadian Single Award (September 2016) combines physical singles, downloads and on-demand streams. Sources: Canadian Encyclopedia (2024a), Music Canada (2026), and RIAA (2026a,b).

point-of-sale terminals in about 2,500 outlets. Eligibility rules on price, track count and running time are set by a Chart Supervisory Committee (OCC, 2023).

D.1.5 Certifications: RIAA and Music Canada

A certification is issued on application by the rights holder, normally the label, who supplies sales documentation and pays for a third-party audit against the award threshold (Music Canada, 2026; RIAA, 2026b). In the physical era both Music Canada and the RIAA certify on net shipments (shipments less returns) (Music Canada, 2026; RIAA, 2026b). Table D2 gives the thresholds.

The retained data comprise 31,468 Music Canada awards dated 1 August 1975 to 24 March 2026, 19.9% of them carrying the source’s Canadian-artist flag, and 56,802 RIAA tier-rows for 29,388 distinct records, dated 14 March 1958 to 20 March 2026. The RIAA publishes one row per award level per record; its unit figure is the record’s current highest-tier total, repeated on every row. Unit estimates in the constructed data convert award tiers using the current threshold schedule.

D.2 Construction

D.2.1 Catalogue and treatment variable

Population All 2,798,335 artist entities in the MusicBrainz dump are collected, 44,946 (1.6%) of them with Canada as country; no country restriction is imposed. Every release dated 1900–2010 whose primary credited artist is in this population is expanded to one row per track. A parallel index of 2,657,525 works records composer, lyricist and writer relations and their nationalities.

Genre exclusion Classical is identified by orchestra or choir artist type, classical work forms (sonata, concerto, symphony, opera, . . .), a composer born before 1900, Latin work language, and classical genre tags; jazz by a curated list of core-jazz tags. Tags match exactly, not as substrings (so “baroque pop”, “rock opera” and “jazz-funk” do not match), and a tag classifies an artist only if it is a dominant share of the artist’s tags. The filters exclude 23,974 artists as classical and 15,822 as jazz. Audiobooks, spoken word, interviews and field recordings are excluded.

Discogs enrichment The catalogues are linked through MusicBrainz’s editor-curated Discogs URL relations, which cover 1,206,271 artists (43.1%). Sleeve credits on linked artists’ Discogs releases are classified into composer, lyricist and writer roles, with nationality resolved through the same link; release country supplies a production-location signal, and genre and style fields supply the genre taxonomy. Signals attach to MusicBrainz tracks on normalised title within artist, with durations within three seconds.

Scoring MAPL is scored per track: two of four components, or, for tracks released from 1992, the collaboration path (an A or P component, plus at least half the composer credits and half the lyricist credits Canadian). Nationality shares are headcounts of credited persons; registered royalty splits are not published. A component with no credit information is scored as not satisfied.

Deduplication Tracks are reduced to one row per artist-recording, dated to the earliest release on which the recording appears. Recordings are then merged into songs where normalised titles match and durations agree within three seconds. Remaster, deluxe and bonus-track annotations are stripped before comparison; “live”, “acoustic”, “demo”, “remix” and “edit” are kept, so alternate versions stay separate. This yields 14,212,647 songs. The song-level measure (`rec_`) is the primary one; a release-appearance measure (`rt_`) is also carried. The correlation of the two cumulative shares is 0.995.

Artist identity Distinct acts sharing a name are kept separate. Duplicate entities for one act are merged when one entity holds at least 90% of the cluster’s combined catalogue and the entities agree on name, sort name, country and type. Output of a joint act named after a member (“Bob Marley & The Wailers”) is also attributed to that member, with a flag. The build merges 520 duplicate identifiers into 500 entities and applies 11,400 such attributions.

Component coverage Table D3 reports the share of songs scored on, and satisfying, each component. 0.1% of songs have a direct recording-location relation; the P component otherwise comes from label country and Discogs release country. 19.6% of songs link to a work. Among songs by Canadian artists, 44.7% have exactly two components scored and 24.4% only one. 90.1% of eligible songs are credited to an artist whose country is Canada.

Artist-year panel Scores are aggregated to credited artist $\times$ year for 1994–2010, balanced by filling gap years from each artist’s debut: 3,860,168 artist-years for 377,814 artists, 714,000 of them with releases. The treatment variable is the cumulative share of an artist’s catalogue up to year t that is MAPL-eligible. Among

Table D3: MAPL component coverage in the constructed catalogue

Component	All songs		Canadian artists		Source of score
	Scored	=1	Scored	=1	
M — Music (composer)	28.6	0.5	31.1	19.7	MB composer relation 12.6; writer proxy 4.9; Discogs 10.8
A — Artist	72.4	2.0	100.0	100.0	Artist country; band-member fallback
P — Production	77.4	1.5	72.8	29.3	Label country 65.7; Discogs release country 11.4; direct location 0.1
L — Lyrics (lyricist)	23.3	0.4	26.5	15.8	MB lyricist relation 8.8; writer proxy 4.8; Discogs 9.4
MAPL-eligible	0.9		42.4		Two-of-four, or post-1991 collaboration path
Linked to a work	19.6		19.0		Recording → work performance relation

Notes: Percentages of songs in the deduplicated catalogue ($N = 14,212,647$; $N = 276,772$ for artists whose MusicBrainz country is Canada). “Scored”: non-missing value for the component. “=1”: component satisfied. Source shares in the last column are percentages of all songs and sum to “All songs / Scored” up to rounding and to instrumental works, for which L does not apply.

Canadian artists it has mean 0.395 and median 0.258; among US and UK artists it is 0.001 or less. 31.2% of Canadian artists show any within-artist change in it. Artists releasing new material in 1994–1998 are flagged (82,761 artists, 21.9%), not filtered.

D.2.2 Digitising RPM

Index The page carrying each chart in each issue was hand-coded from the scans: 4,107 (issue date, chart, page) records for 1964–2000, of which 2,226 chart pages fall in 1994–2000.

Extraction Each indexed page is rendered at 300 dpi and passed to a vision-language model with a chart-specific prompt. The model returns one record per entry: this-week rank, last-week rank, weeks on chart, the text lines verbatim, and the two Canadian-content markers printed by *RPM*. The prompt first requires the page’s printed heading to match a known variant of the target chart name, and returns nothing otherwise. Decoding is deterministic. Artist and title are parsed from the verbatim lines afterwards (Section D.2.3). The printed Canadian-content markers are captured but not used.

Output All 2,226 pages returned parseable output: 147,698 entries across 324 issues from 10 January 1994 to 6 November 2000. Three pages returned no entries and were re-read after correcting the index; three chart-page misidentifications found downstream were corrected. In 99.5% of chart-issues the number of entries equals the highest printed rank. 105 chart-issues are flagged by an outlier rule (catalogue match rate low relative to a rolling median for the chart).

Table D4 describes the panel. *RPM 100 Hit Tracks*, *Top Albums* and *Country Tracks* keep a constant depth over the window; Adult Contemporary, Dance and Country Albums change depth. Alternative ends in December 1998, Rock begins in April 1999, and Urban covers ten months of 1998. Each year has 44 to 49

Table D4: RPM charts: coverage, depth, and match rates

Chart	Issues	Span	Depth	Entries	Artist	Title
RPM 100 Hit Tracks	323	1994:01–2000:11	100	32,301	87.8	79.2
Top Albums	322	1994:01–2000:11	100	32,300	93.8	77.8
Country Tracks	323	1994:01–2000:11	100	32,235	87.3	74.4
Adult Contemporary	323	1994:01–2000:11	40 / 60 / 100	21,063	89.6	81.4
Country Albums	322	1994:01–2000:11	33 / 40 / 30	12,369	95.3	72.6
Dance	320	1994:01–2000:10	10 / 30 / 40	8,760	85.2	69.9
Alternative	169	1995:06–1998:12	30	5,070	95.2	89.2
Rock	79	1999:04–2000:11	30	2,370	95.4	90.4
Urban	41	1998:02–1998:12	30	1,230	76.7	64.9
Total	324	1994:01–2000:11		147,698	89.8	77.5

Notes: “Issues”: distinct issue dates on which the chart was read; the total row counts distinct issues over all charts. Depth: number of ranked positions (Adult Contemporary 40 in 1994, 60 from 1995, 100 from mid-1999; Dance 10 in 1994, 30 from 1995, 40 in 2000; Country Albums 33 in 1994, 40 from 1995, 30 in 2000). “Artist” and “Title”: percentages of entries resolved to a catalogue artist and to a catalogue recording or release group, excluding “various artists” entries (7,237, 4.9% of the total: 6,205 on Top Albums, 1,032 on Country Albums).

issues, reflecting combined and skipped issues; 301 of the 320 intervals between consecutive issues are seven days.

D.2.3 Record linkage

Outcome sources identify records by printed artist name and title; the treatment is keyed to catalogue identifiers. The same procedure links all four outcome sources.

Parsing *RPM*’s verbatim lines are split into artist and title by chart-specific rules: title then “artist – album” on most singles charts, the reverse on album charts, and a slash-delimited artist/album/availability line on the Country chart. A slash preceded by “w” (as in “Lee Ann Womack W/Sons of the Desert”) is not a field separator. “Various artists” is detected in either field; compilation entries are retained and not matched.

Normalisation Names and titles are case-folded and stripped of punctuation and diacritics; titles also lose remaster, edit and version qualifiers. Only the primary credited artist is matched. Secondary credits are split off on w/, feat., featuring, and, for UK data, presents; ampersands and “and” are not split (*AC/DC*, *Earth, Wind & Fire*).

Matching Entries are collapsed to distinct artist names, each resolved against the artist table, then to distinct titles, each resolved within the matched artist’s own catalogue. Scoring is fuzzy string similarity. Each entry is classified as matched, ambiguous, no match, no candidates, or excluded; unresolved entries are kept with their classification.

Ambiguity Same-name candidates (“Drake” resolves to five entities, “Eminem” to three) are resolved by four rules in order: an exact name match beats a tie reached through subset scoring against a different name;

Table D5: Record linkage match rates, by source and period

Source	Format	Entries dated before 2010			Entries dated 2010 or later		
		<i>N</i>	Artist	Title	<i>N</i>	Artist	Title
RPM (Canada)	singles	103,029	88.2	77.9	–		
	albums	37,432	94.3	76.2	–		
Billboard (US)	singles	1,686,884	97.1	80.4	1,476,010	82.9	17.6
	albums	1,522,997	83.8	62.8	968,891	80.8	20.6
Billboard (Canada)	singles	13,500	99.5	95.2	85,000	81.5	11.1
	albums	3,045	87.3	79.5	62,125	81.7	16.1
Official Charts (UK)	singles	348,277	97.3	83.1	236,993	82.6	29.0
	albums	460,350	78.2	69.3	448,341	64.9	23.9
Music Canada	singles	1,201	98.8	89.3	13,745	82.7	12.9
	albums	11,273	89.1	77.5	2,909	86.4	22.4
RIAA (US)	singles	4,266	96.1	88.0	28,361	82.3	20.9
	albums	17,132	94.9	79.3	4,596	86.8	29.5

Notes: “Artist” and “Title”: percentages of source rows resolved to a catalogue artist and to a catalogue recording or release group. The denominator is all rows in the cell, including compilation rows, which are not matched. Titles are matched only within a resolved artist’s catalogue. RPM covers 1994–2000; Billboard and Official Charts cover the full retained archive, 1956/1952–2026. Certifications are dated by certification, not release.

between exact duplicates, the one with the dominant catalogue wins; candidates whose debut postdates the entry are eliminated; and the charted title is searched across surviving candidates’ catalogues, the owner of the best-matching title winning. Remaining ties are left ambiguous. When two recordings in one artist’s catalogue both match a title perfectly, the earliest release is taken.

D.2.4 Match rates

Table D5 reports match rates by source, format, and period, split at 2010. Compilation, cast-recording and soundtrack entries are 25.4% of UK album-chart rows. For entries dated 2010 or later, title match rates are 11–30% and artist match rates 65–87%. No estimate in the paper uses post-2010 outcomes.

As a check on false artist matches, I count artists matched with confidence for whom none of their charted titles is found in their catalogue. In the *RPM* data these are 607 artists and 4,586 entries (3.1% of the panel). The count includes correctly identified artists whose catalogue lacks the charted titles.

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